

Lecture 3 – Friday March 15, 2012

Kinematics of Robots: Position Analysis

Some slides of this lecture are based on Chapter 2 of Saeed Benjamin Niku. *Introduction to Robotics*. 2nd Ed., Wiley, 2011.

Objectives

When you have finished this lecture you should be able to:

- Recognize methods for representation of points, vectors, frames, and transformations by matrices.
- Learn how to represent the motions of a multi-DOF robot in space and how to derive the forward kinematic equations of different robot configurations such as Cartesian, cylindrical, and spherical coordinates.

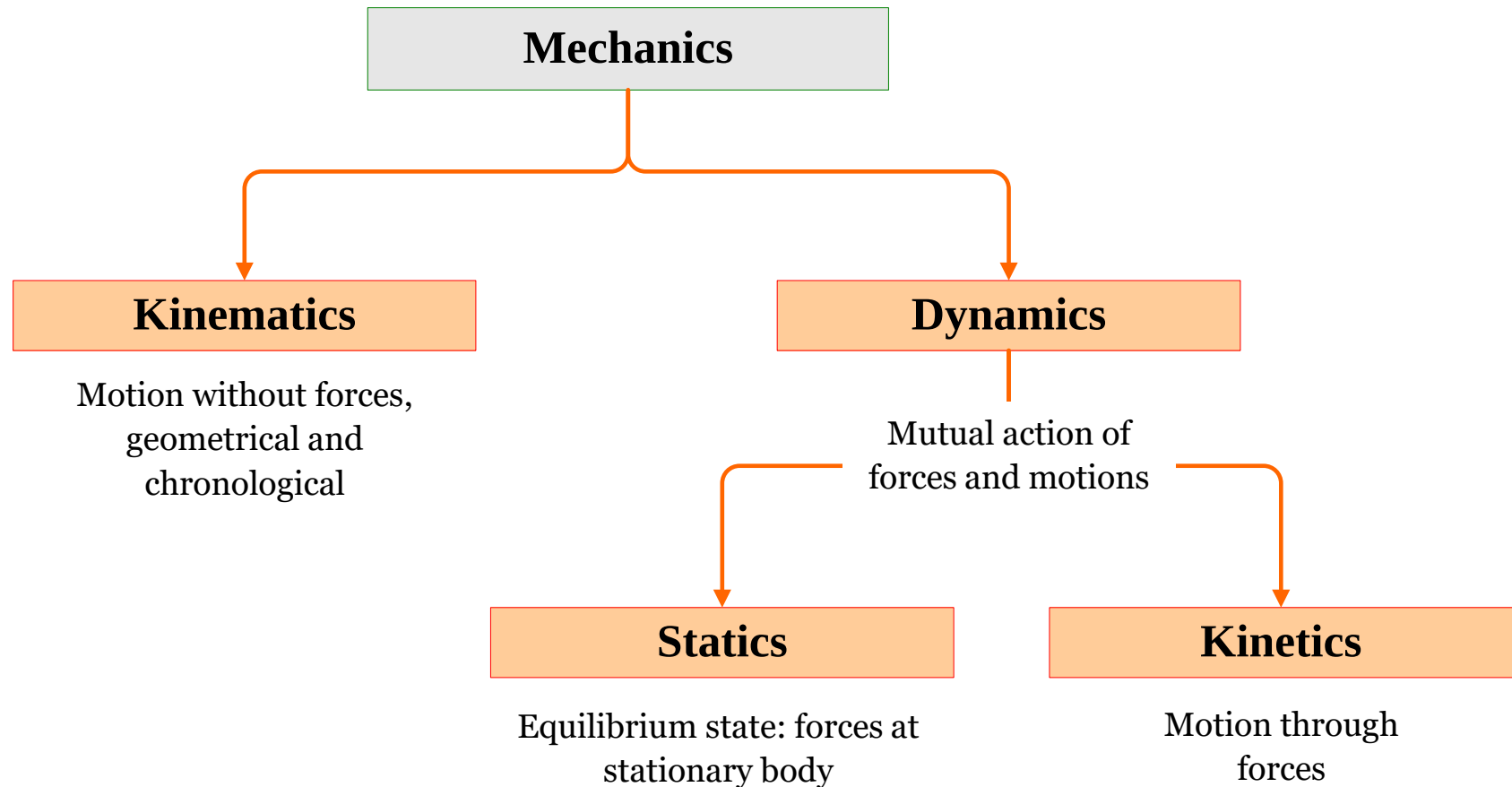
Outline

- Kinematics
- Robots as Mechanisms
- Matrix Representation
- Homogenous Transformation Matrices
- Inverse of Transformation Matrices
- Forward and Inverse Kinematic Equations [For Reading]
- Summary

Outline

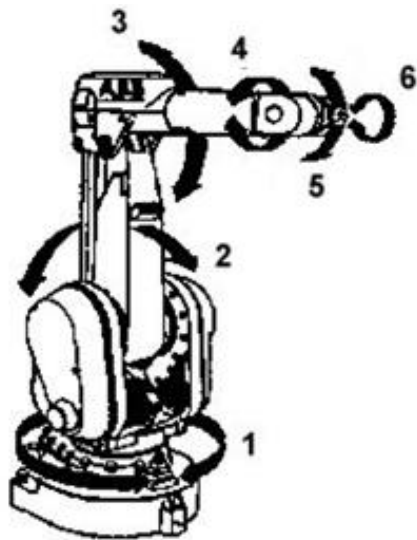
- **Kinematics**
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Kinematics



Kinematics

Kinematics is the relationships between the positions, velocities, and accelerations of the links of a manipulator.



Given: The angle of each joint

Required: The position of end-effector or any point (i.e. its coordinates)

$(x, y, z, \alpha, \beta, \gamma)$

Forward Kinematics



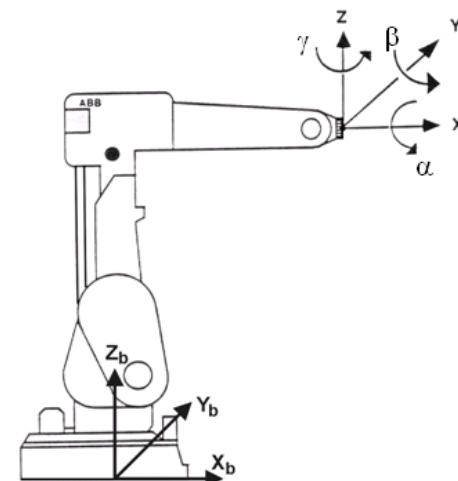
Inverse Kinematics



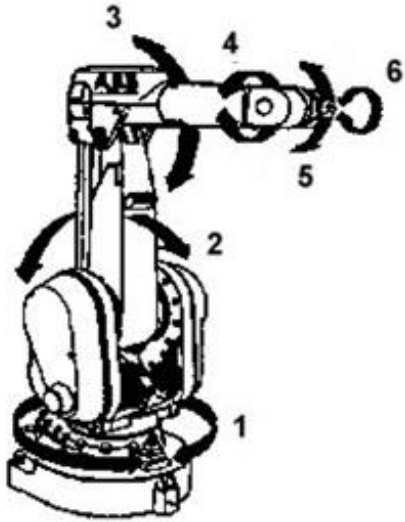
Given: The position of end-effector or some point in the robot workspace

Required: The angles of each joint needed to obtain that position

$(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$



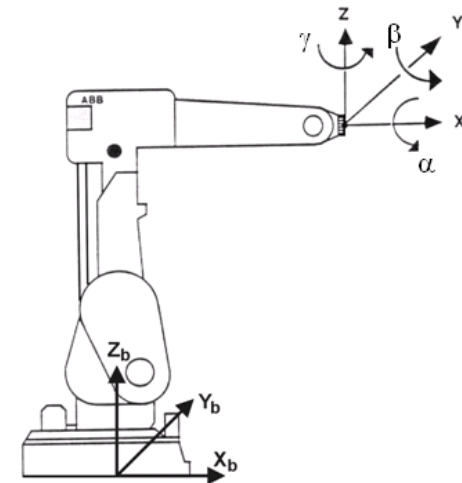
Kinematics



Forward Kinematics



Inverse Kinematics



$$\theta_k = f_k(x, y, z, \alpha, \beta, \gamma)$$

$$K = 1, \dots, n \quad (n \text{ is DOF} = 6)$$

$$x = f_x(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$$

$$y = f_y(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$$

$$z = f_z(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$$

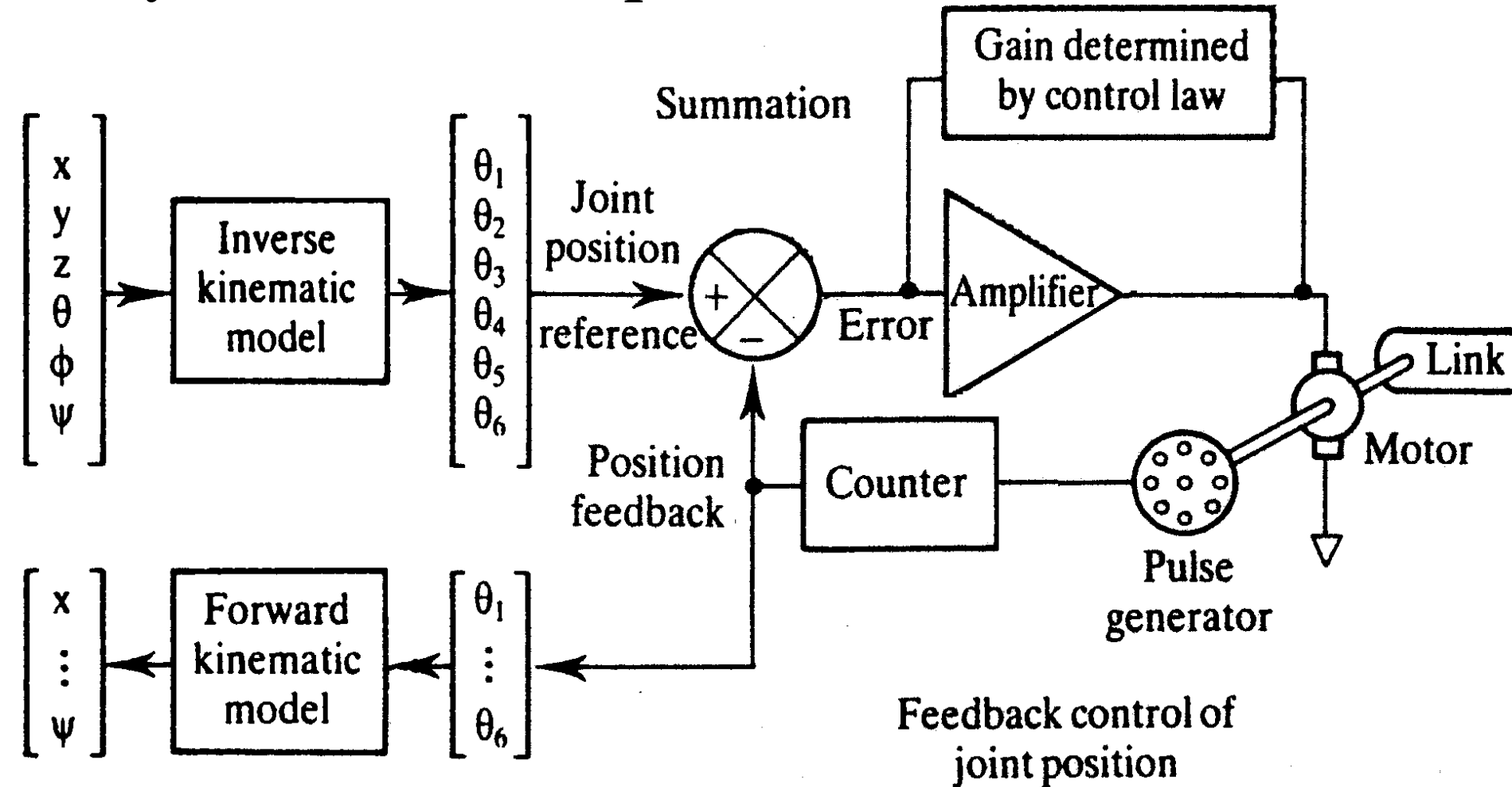
$$\alpha = f_\alpha(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$$

$$\beta = f_\beta(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$$

$$\gamma = f_\gamma(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$$

Kinematics

- Why Kinematics is important?



Outline

- Kinematics
- **Robots as Mechanisms**
- Matrix Representation
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- Summary

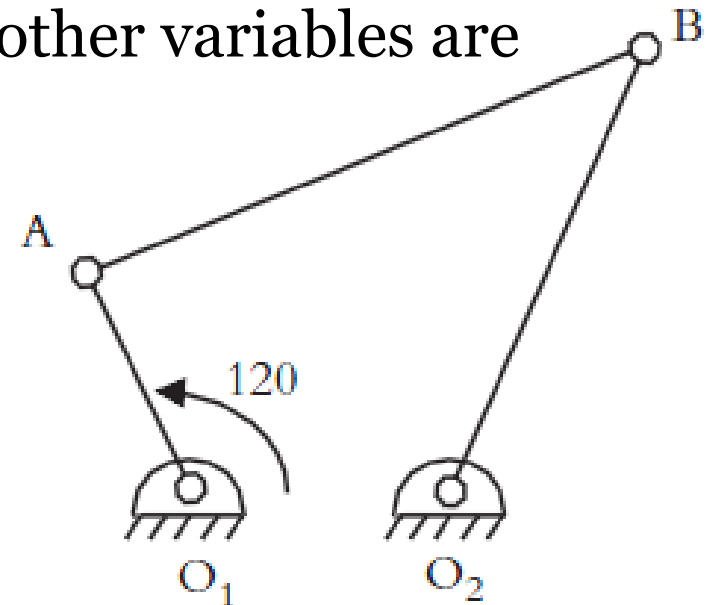
Robots as Mechanisms

Manipulator-type robots are multi-degree-of-freedom (DOF), three-dimensional, open loop, chain mechanisms

- **Multi-DOF:** means that robots possess many joints, allowing them to move freely within their envelope.

In a 1-DOF system, when the variable is set to a particular value, the mechanism is totally set and all its other variables are known.

Example: in the 1-DOF 4-bar mechanism, when the crank is set to 120° , the angles of the coupler link and the rocker arm are also known.

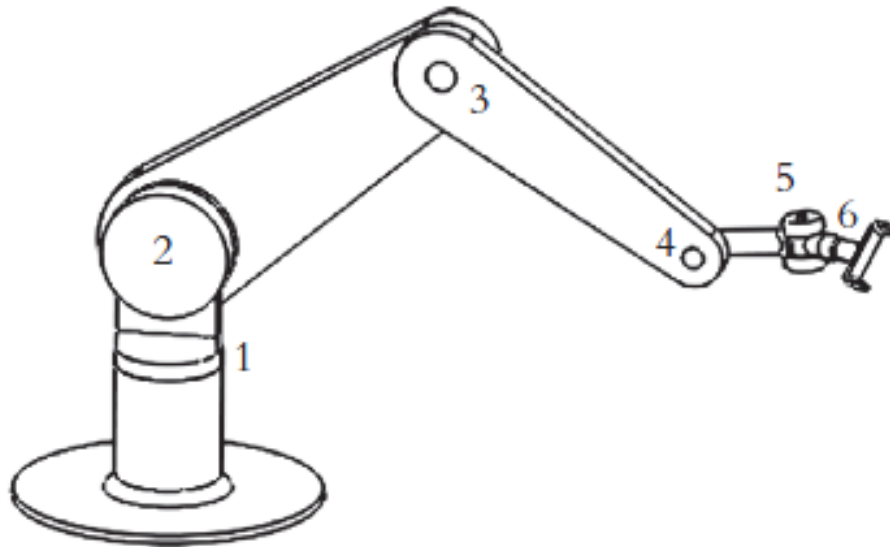


A 1-DOF closed-loop 4-bar mechanism

Robots as Mechanisms

- **Multi-DOF (cont'd):**

In a multi-DOF mechanism, all input variables must be individually defined in order to know the remaining parameters. Robots are multi-DOF machines, where **each joint variable** must be known in order to determine the location of the **robot's hand**.



Robots as Mechanisms

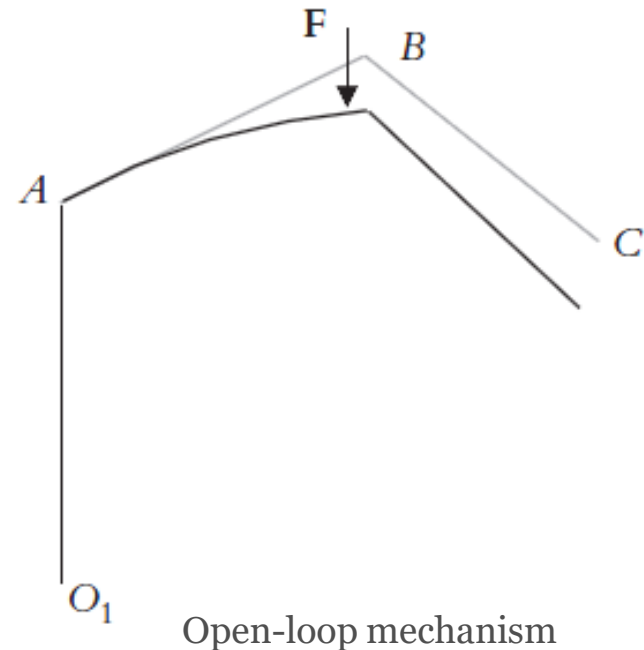
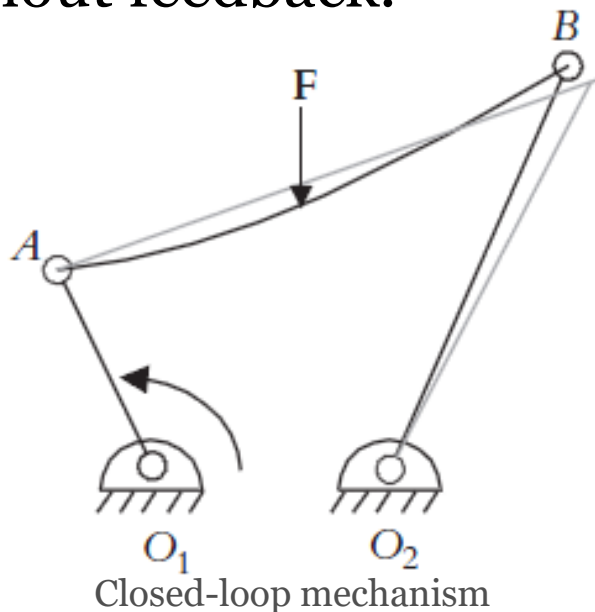
- **Dimensions:**

Robots are three-dimensional machines if they are to move in space. Although it is possible to have a two-dimensional multi-DOF robot, they are not common (or useful).

Robots as Mechanisms

- **Closed-loop/open-loop mechanisms:**

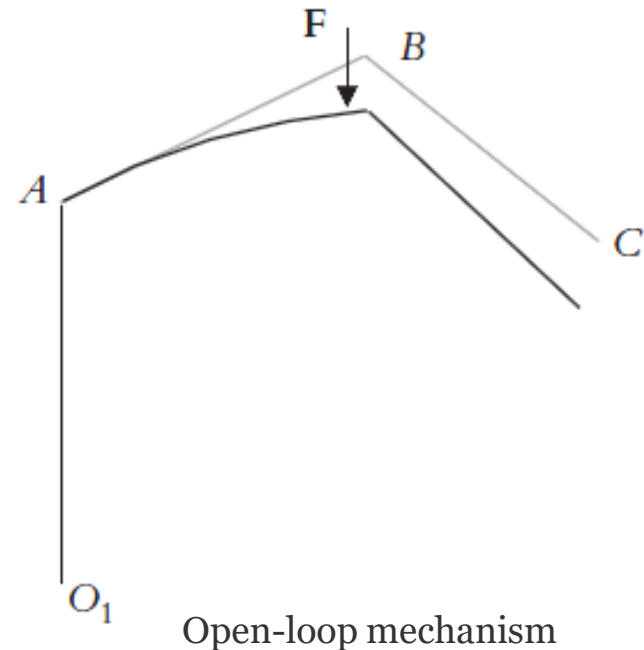
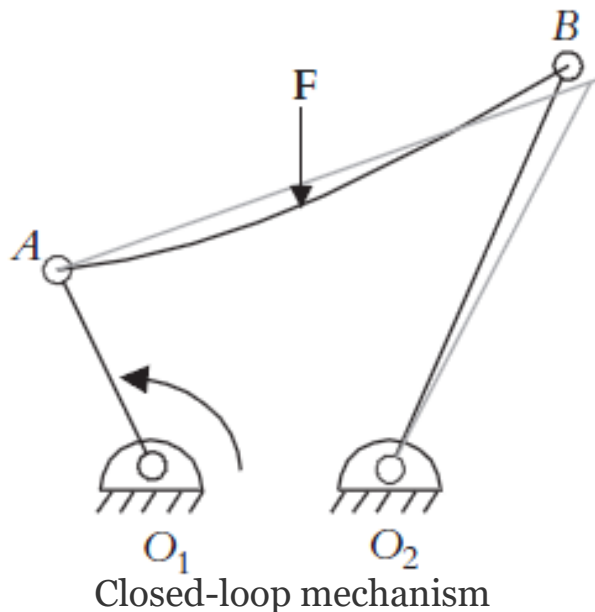
Robots are open-loop mechanisms. Unlike mechanisms that are closed-loop (e.g., 4-bar mechanisms), even if all joint variables are set to particular values, there is no guarantee that the hand will be at the given location. This is because deflections in any joint or link will change the location of all subsequent links without feedback.



Robots as Mechanisms

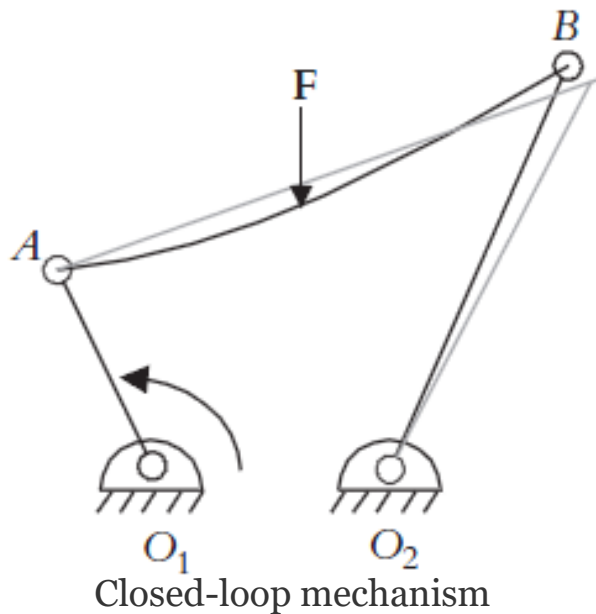
- **Closed-loop/open-loop mechanisms:**

Example: when link AB deflects as a result of load F, link BO₂ will also move; therefore, the deflection can be detected. In an open-loop system such as the robot, the deflections will move all succeeding members without any feedback.

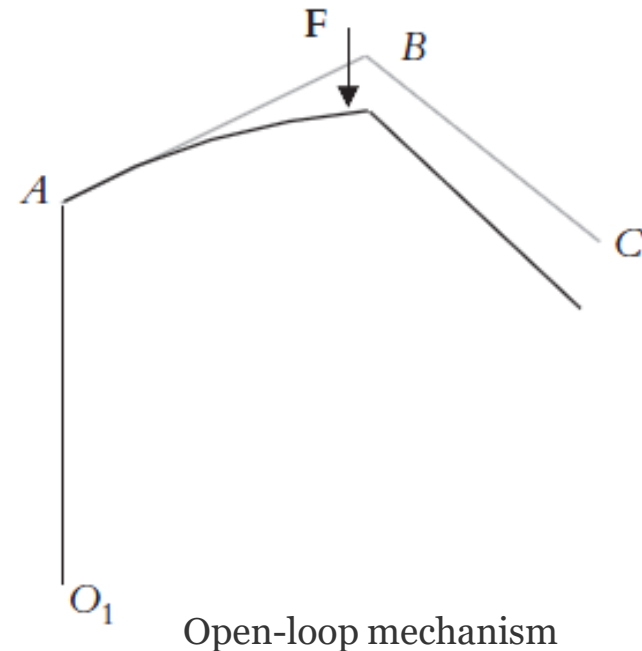


Robots as Mechanisms

- Closed-loop/open-loop mechanisms:



$$\overline{O_1A} + \overline{AB} = \overline{O_1O_2} + \overline{O_2B}$$



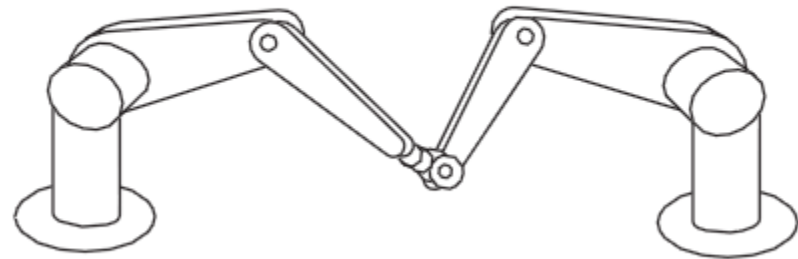
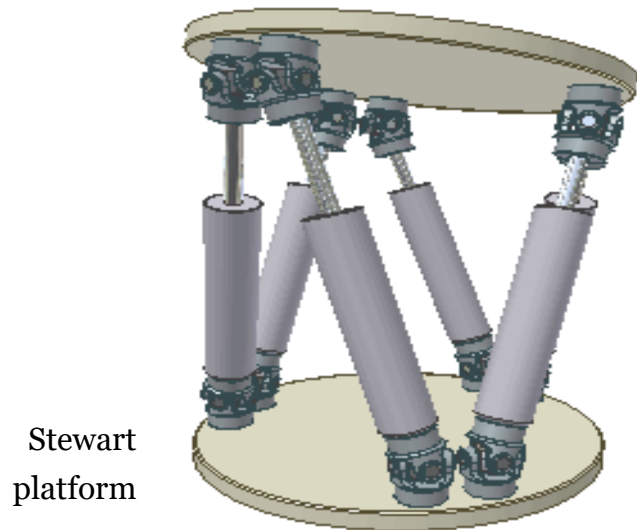
$$\overline{O_1A} + \overline{AB} + \overline{BC} = \overline{O_1C}$$

In *open-loop mechanism*, unless O_1C is measured by feedback, the change will not be known.

Robots as Mechanisms

- **Closed-loop/open-loop mechanisms:**

Parallel manipulators are based on closed-loop parallel architecture. The trade-off is much-reduced range of motions and workspace.



Possible parallel manipulator configurations

Outline

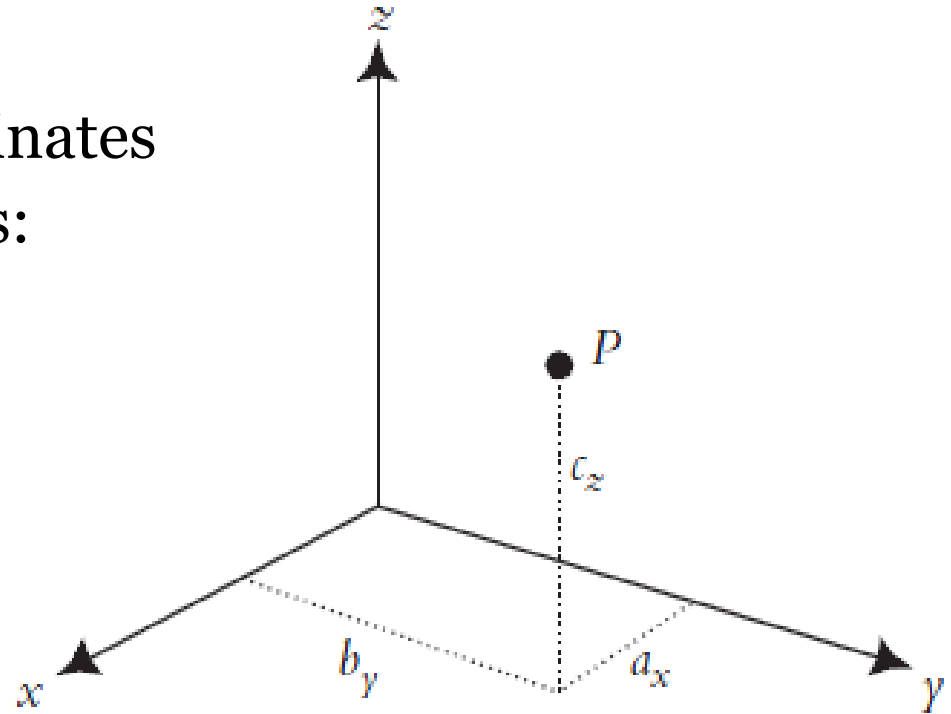
- Kinematics
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- **Matrix Representation**
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Matrix Representation

- **Representation of a Point in Space**

A point P in space can be represented by its three coordinates relative to a reference frame as:

$$P = a_x i + b_y j + c_z k$$



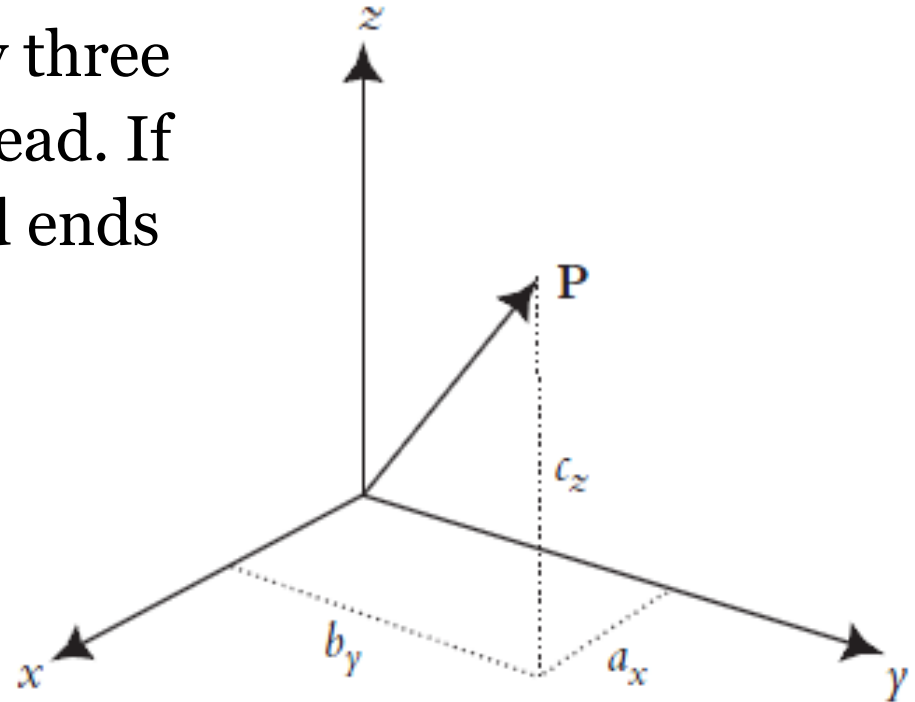
Obviously, other coordinate representations such as cylindrical or spherical can also be used to describe the location of a point in space.

Matrix Representation

• Representation of a Vector in Space

A vector can be represented by three coordinates of its tail and its head. If the vector starts at point A and ends at point B.

$$\begin{aligned} P_{AB} = & (B_x - A_x)i \\ & + (B_y - A_y)j \\ & + (B_z - A_z)k \end{aligned}$$



Specifically, if the vector starts at the origin

$$P = a_x i + b_x j + c_z k$$

where a_x , b_y and c_x are the three components of the vector in the reference frame.

Matrix Representation

- **Representation of a Vector in Space**

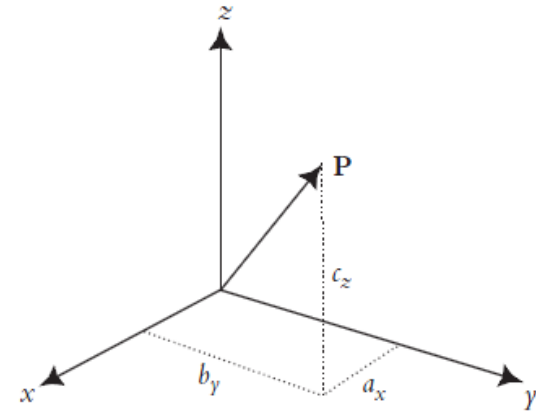
In matrix form:

$$P = \begin{bmatrix} a_x \\ b_y \\ c_z \end{bmatrix}$$

This representation can be slightly modified to also include a scale factor w

$$P = \begin{bmatrix} P_x \\ P_y \\ P_z \\ w \end{bmatrix}$$

where $P_x = wa_x$, $P_y = wb_y$, $P_z = wc_z$.



Matrix Representation

- **Representation of a Vector in Space**

Example: A vector P is 5 units long and is in the direction of a unit vector q described below. Express the vector in matrix form.

$$q = \begin{bmatrix} 0.371 \\ 0.557 \\ q_z \\ 1 \end{bmatrix}$$

Solution: the unit vector's length must be 1. Therefore.

$$\lambda = \sqrt{q_x^2 + q_y^2 + q_z^2} = \sqrt{0.138 + 0.310 + q_z^2} = 1 \rightarrow q_z = 0.743$$

$$q_{unit} = \begin{bmatrix} 0.371 \\ 0.557 \\ 0.743 \\ 1 \end{bmatrix} \quad \text{and} \quad p = q_{unit} \times 5 = \begin{bmatrix} 1.855 \\ 2.785 \\ 3.715 \\ 1 \end{bmatrix}$$

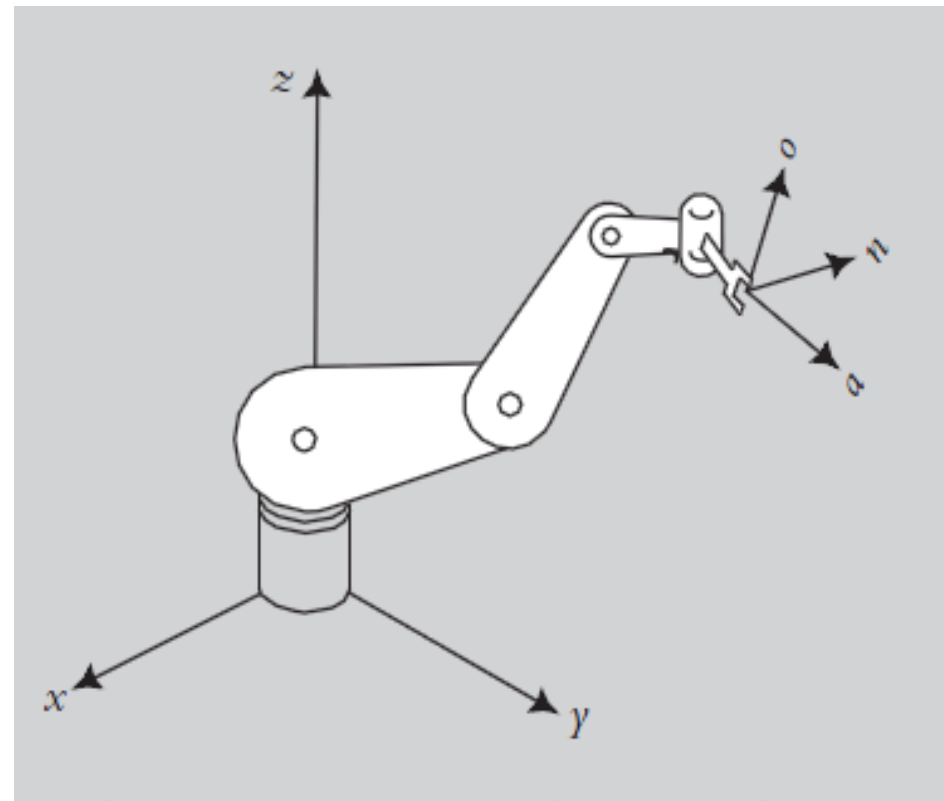
Matrix Representation

- **Representation of a Frame**

Fixed Universe reference frame $\gg F_{x;y;z}$

Another (moving) frames $\gg F_{n;o;a}$ relative to the reference frame

Note: n, o and a stands for normal, orientation, and approach.

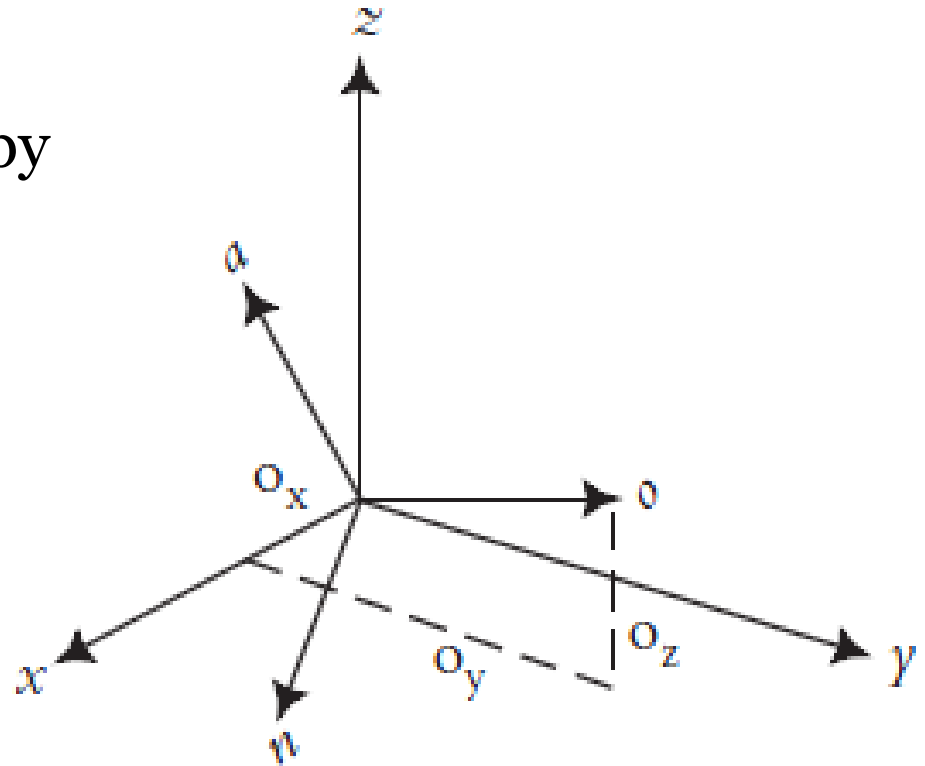


Matrix Representation

- **A Frame at the origin of the reference frame**

The moving frame as $F_{n;o;a}$ with normal, orientation, and approach axes is represented by the following matrix:

$$F = \begin{bmatrix} n_x & o_x & a_x \\ n_y & o_y & a_y \\ n_z & o_z & a_z \end{bmatrix}$$



Representation of a frame at the origin
of the reference frame

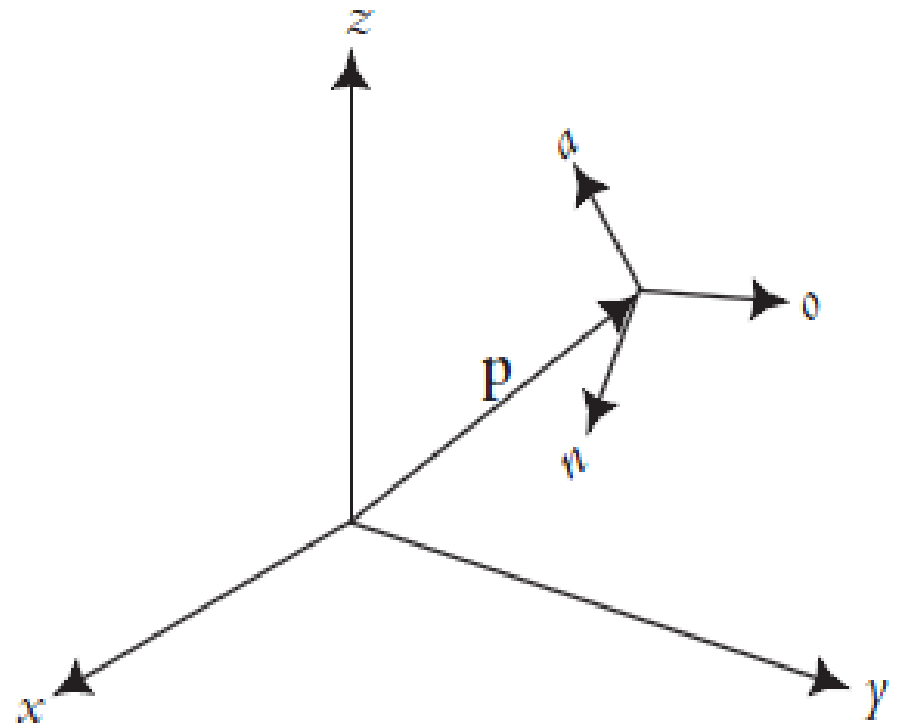
Matrix Representation

- **A Frame relative to a fixed reference frame**

The frame can be expressed by three vectors describing its directional unit vectors and a fourth vector describing its location.

$$F = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note: The frame may also be represented by a 3x4 matrix without the scale factors, but it is not common. Adding the fourth row of scale factors to the matrix makes it a 4x4 or homogeneous matrix.



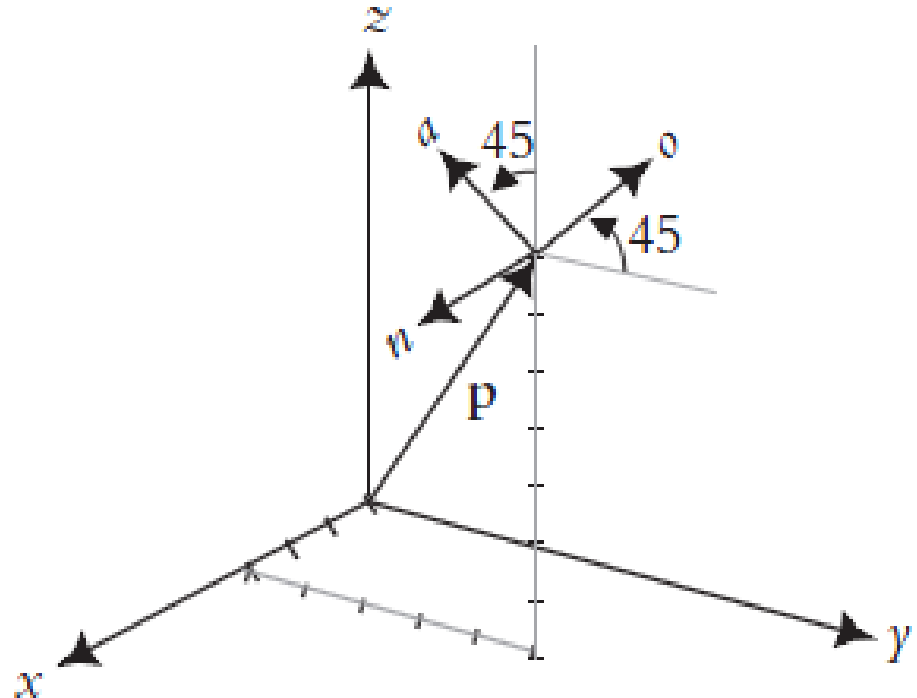
Representation of a frame in a frame

Matrix Representation

- **A Frame relative to a fixed reference frame**

Example: The frame F is located at 3,5,7 units, with its n-axis parallel to x, its o-axis at 45° relative to the y-axis, and its a-axis at 45° relative to the z-axis. The frame can be described by:

$$F = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 0.707 & -0.707 & 5 \\ 0 & 0.707 & 0.707 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

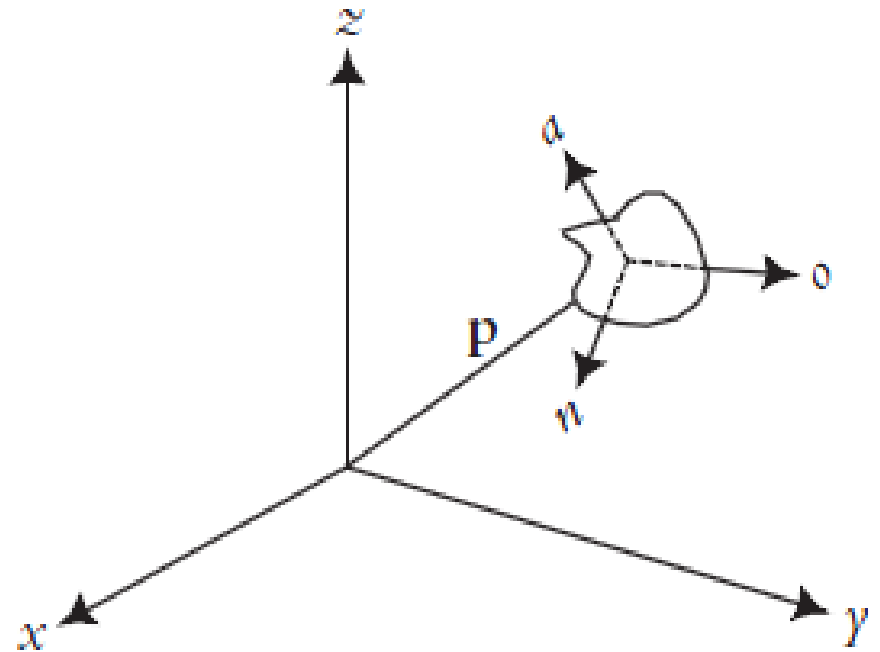


Matrix Representation

- **Representation of a Rigid Body**

An object can be represented in space by attaching a frame to it and representing the frame. Since the object is permanently attached to this frame, its position and orientation relative to the frame is always known.

$$F = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Representation of an object in space

Matrix Representation

- **Representation of a Rigid Body**

A **point** in space has only **three degrees of freedom**; it can only move along the three reference axes.

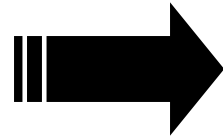
However, a **rigid body** in space has **six degrees of freedom**, meaning that not only can it move along x-, y-, and z-axes, it can also rotate about these three axes.

Consequently, all that is needed to completely define an object in space is **six pieces of information** describing the location of the origin of the object in the reference frame and its orientation about the three axes.

Matrix Representation

• Representation of a Rigid Body

$$F = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



12 pieces of information:

- 9 for orientation
- 3 for position

Note: this excludes the scale factors on the last row of the matrix because they do not add to this information.

Obviously, there must be some **constraints** present in this representation to limit the above to six.

Matrix Representation

- **Representation of a Rigid Body**

We need **6 constraint equations** to reduce the above from twelve to six.

The constraints come from the **known characteristics of a frame** that have not been used yet, that:

- The three unit vectors n , o , a are **mutually perpendicular**, and
- Each **unit vector's length**, represented by its directional cosines, must be equal to 1.

Matrix Representation

- Representation of a Rigid Body

Six constraint equations:

1: $\mathbf{n} \cdot \mathbf{o} = 0$ (dot-product of $\mathbf{n}$ and $\mathbf{o}$ vectors must be zero)

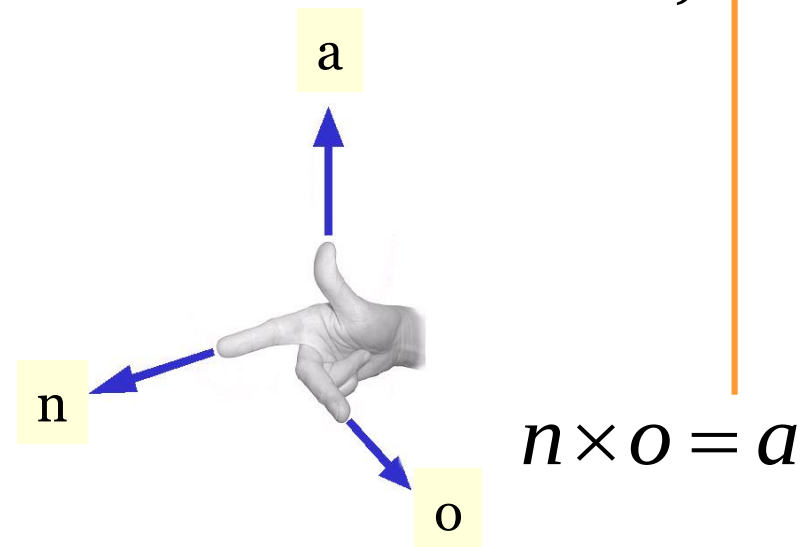
2: $\mathbf{n} \cdot \mathbf{a} = 0$

3: $\mathbf{a} \cdot \mathbf{o} = 0$

4: $|\mathbf{n}| = 1$ (the magnitude of the length of the vector must be 1)

5: $|\mathbf{o}| = 1$

6: $|\mathbf{a}| = 1$



Matrix Representation

- **Representation of a Rigid Body**

Example: For the following frame, find the values of the missing elements and complete the matrix representation of the frame:

$$F = \begin{bmatrix} ? & 0 & ? & 5 \\ 0.707 & ? & ? & 3 \\ ? & ? & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution: Obviously, the 5,3,2 values representing the position of the origin of the frame do not affect the constraint equations. Please notice that only 3 values for directional vectors are given. This is all that is needed.

Matrix Representation

- Representation of a Rigid Body

Example (cont'd):

$$n_x o_x + n_y o_y + n_z o_z = 0 \quad \text{or} \quad n_x(0) + 0.707(o_y) + n_z(o_z) = 0$$

$$n_x a_x + n_y a_y + n_z a_z = 0 \quad \text{or} \quad n_x(a_x) + 0.707(a_y) + n_z(0) = 0$$

$$a_x o_x + a_y o_y + a_z o_z = 0 \quad \text{or} \quad a_x(0) + a_y(o_y) + 0(o_z) = 0$$

$$n_x^2 + n_y^2 + n_z^2 = 1 \quad \text{or} \quad n_x^2 + 0.707^2 + n_z^2 = 1$$

$$o_x^2 + o_y^2 + o_z^2 = 1 \quad \text{or} \quad 0^2 + o_y^2 + o_z^2 = 1$$

$$a_x^2 + a_y^2 + a_z^2 = 1 \quad \text{or} \quad a_x^2 + a_y^2 + 0^2 = 1$$

Simplifying these equations yields:

Matrix Representation

- Representation of a Rigid Body

Example (cont'd):

$$0.707 \, o_y + n_z o_z = 0$$

$$n_x a_x + 0.707 \, a_y = 0$$

$$a_y o_y = 0$$

$$n_x^2 + n_z^2 = 0.5$$

$$o_y^2 + o_z^2 = 1$$

$$a_x^2 + a_y^2 = 1$$

Solving these six equations will yield

$$n_x = \pm 0.707; n_z = 0; o_y = 0; o_z = 1;$$

$$a_x = \pm 0.707, \text{ and } a_y = -0.707.$$

Notice that both n_x and a_x must have the same sign. The reason for multiple solutions is that with the given parameters, it is possible to have two sets of mutually perpendicular vectors in opposite directions.

Matrix Representation

- Representation of a Rigid Body

Example (cont'd):

$$F = \begin{bmatrix} 0.707 & 0 & 0.707 & 5 \\ 0.707 & 0 & -0.707 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{or} \quad F = \begin{bmatrix} -0.707 & 0 & -0.707 & 5 \\ 0.707 & 0 & -0.707 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Both matrices satisfy all the requirements set by the constraint equations. It is important to realize that the values representing the three direction vectors are not arbitrary but bound by these equations. Therefore, you may not randomly use any desired values in the matrix.

Note: The same problem may be solved using $\mathbf{n}\mathbf{x}\mathbf{o}=\mathbf{a}$

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Homogenous Transformation Matrices

- **Homogenous Matrix**

$$T(4 \times 4) = \begin{pmatrix} \text{Rotation}(3 \times 3) & \text{Translation}(3 \times 1) \\ \text{Deformation}(1 \times 3) & \text{Scale}(1 \times 1) \end{pmatrix}$$

In **robotics**, we are normally interested in rotation and translation transformations, as we are dealing with **solid physical objects** not computer generated images. Also, because the scale factor is 1, the 4 dimensional transformation maps directly into three dimensions.

$$T(4 \times 4) = \begin{pmatrix} \text{Rotation}(3 \times 3) & \text{Translation}(3 \times 1) \\ 0 & 1 \end{pmatrix}$$

Homogenous Transformation Matrices

- **Representation of Transformation**

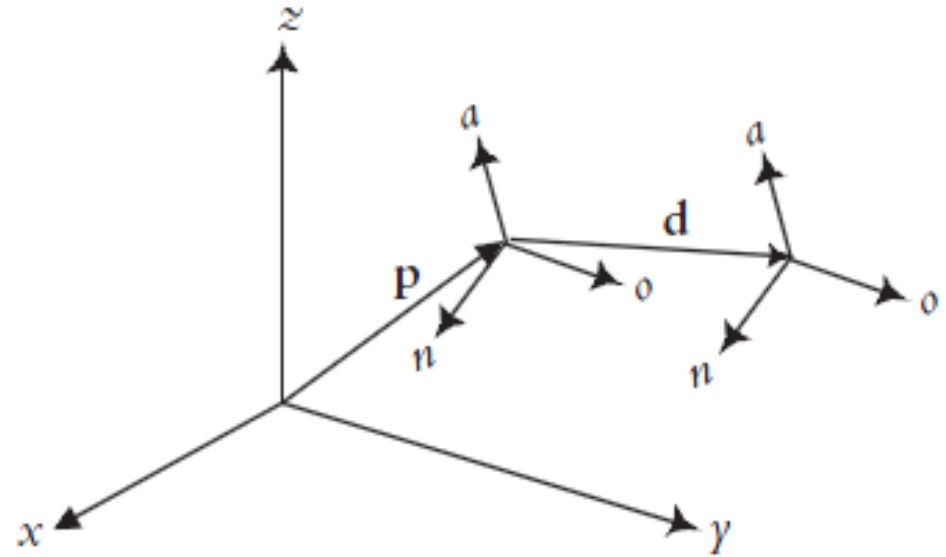
A transformation is defined as making a movement in space. A transformation may be in one of the following forms:

- A pure translation
- A pure rotation about an axis
- A combination of translations and/or rotations

Homogenous Transformation Matrices

• Representation of a Pure Translation

$$T = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$F_{new} = Trans(d_x, d_y, d_z) \times F_{old}$$

$$F_{new} = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} n_x & o_x & a_x & p_x + d_x \\ n_y & o_y & a_y & p_y + d_y \\ n_z & o_z & a_z & p_z + d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogenous Transformation Matrices

- **Representation of a Pure Translation**

Example: A frame F has been moved 10 units along the y-axis and 5 units along the z-axis of the reference frame. Find the new location of the frame.

$$F = \begin{bmatrix} 0.527 & -0.574 & 0.628 & 5 \\ 0.369 & 0.819 & 0.439 & 3 \\ -0.766 & 0 & 0.643 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogenous Transformation Matrices

- Representation of a Pure Translation

Solution:

$$F_{new} = Trans(d_x, d_y, d_z) \times F_{old} = Trans(0, 10, 5) \times F_{old}$$

and

$$F_{new} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0.527 & -0.574 & 0.628 & 5 \\ 0.369 & 0.819 & 0.439 & 3 \\ -0.766 & 0 & 0.643 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 0.527 & -0.574 & 0.628 & 5 \\ 0.369 & 0.819 & 0.439 & 13 \\ -0.766 & 0 & 0.643 & 13 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogenous Transformation Matrices

• Representation of a Pure Rotation about an Axis

If the frame is rotating about x-axis, the value of p_x does not change,

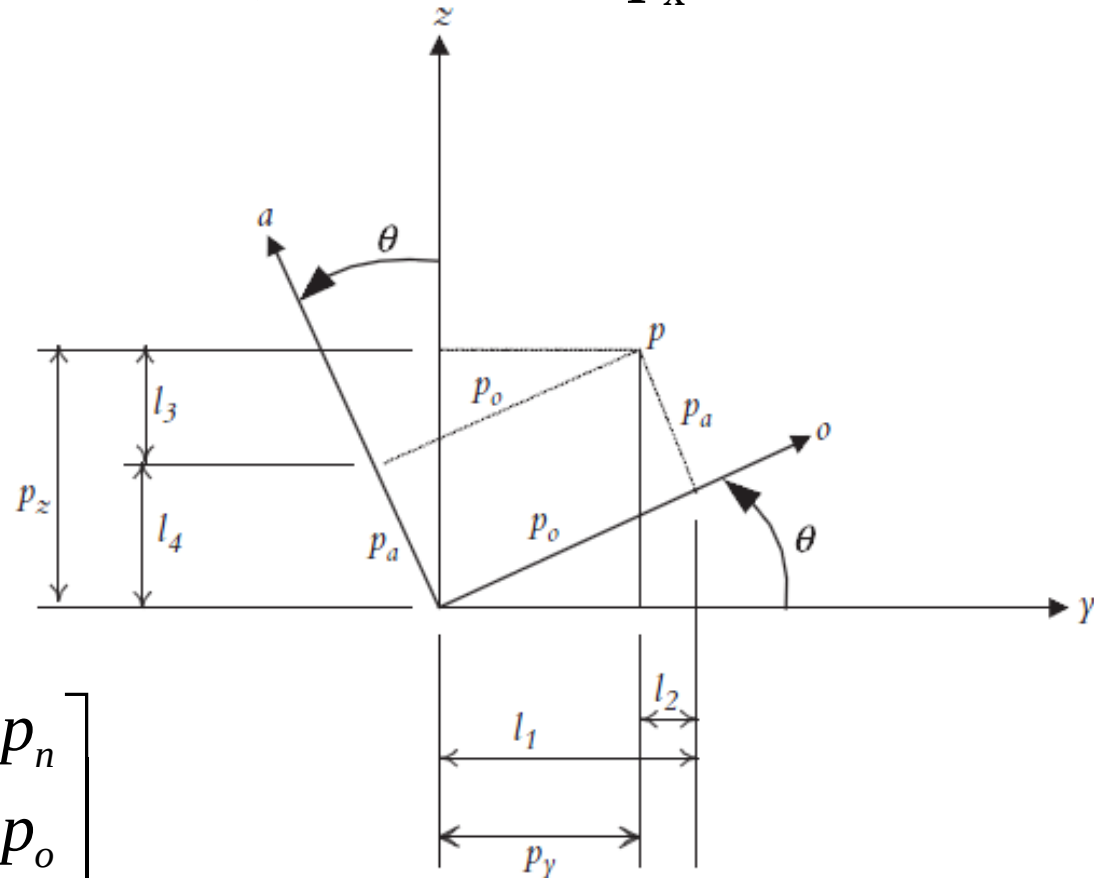
but the values of p_y and p_z do change.

$$p_x = p_n$$

$$p_y = l_1 - l_2 = p_o \cos \theta - p_a \sin \theta$$

$$p_z = l_3 + l_4 = p_o \sin \theta + p_a \cos \theta$$

$$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} p_n \\ p_o \\ p_a \end{bmatrix}$$



Coordinates of a point relative to the reference frame and rotating frame as viewed from the x-axis.

Homogenous Transformation Matrices

- **Representation of a Pure Rotation about an Axis**

Rotation about y-axis:

$$Rot(y, \theta) = \begin{bmatrix} C\theta & 0 & S\theta \\ 0 & 1 & 0 \\ -S\theta & 0 & C\theta \end{bmatrix}$$

Rotation about z-axis:

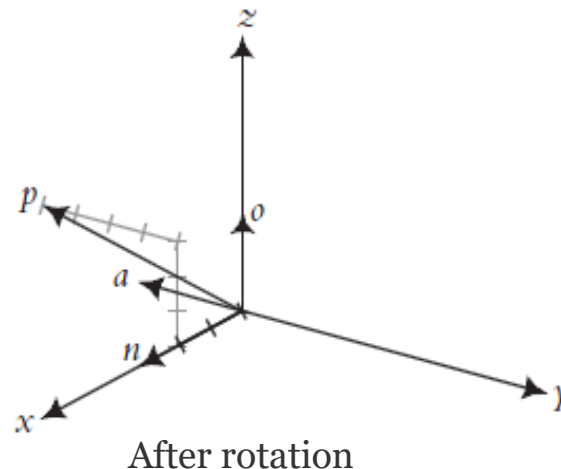
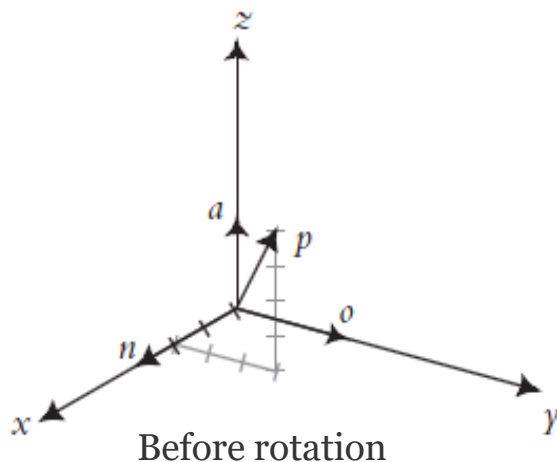
$$Rot(z, \theta) = \begin{bmatrix} C\theta & -S\theta & 0 \\ S\theta & C\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Homogenous Transformation Matrices

- **Representation of a Pure Rotation about an Axis**

Example: A point $p(2,3,4)^T$ is attached to a rotating frame. The frame rotates 90° about the x-axis of the reference frame. Find the coordinates of the point relative to the reference frame after the rotation, and verify the result graphically.

$$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & C\theta & -S\theta \\ 0 & S\theta & C\theta \end{bmatrix} \times \begin{bmatrix} p_n \\ p_o \\ p_a \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 3 \end{bmatrix}$$



Homogenous Transformation Matrices

- **Representation of Combined Transformations**

1. Rotation of α degrees about the x-axis,
2. Followed by a translation of $[l_1, l_2, l_3]$ (relative to the x-, y-, and z-axes respectively),
3. Followed by a rotation of β degrees about the y-axis.

$$p_{x,y,z} = Rot(y, \beta) \times Trans(l_1, l_2, l_3) \times Rot(x, \alpha) \times p_{noa}$$



Pre-multiplication

Homogenous Transformation Matrices

- **Representation of Combined Transformations**

Example: A point $p(7,3,1)^T$ is attached to a frame F_{noa} and is subjected to the following transformations. Find the coordinates of the point relative to the reference frame at the conclusion of transformations.

1. Rotation of 90° about the z-axis,
2. Followed by a rotation of 90° about the y-axis,
3. Followed by a translation of $[4,3,7]$.

Homogenous Transformation Matrices

- **Transformations Relative to the Rotating Frame**

Example:

1. A rotation of 90° about the a-axis,
2. Then a translation of $[4,3,7]$ along n-, o-, a-axes
3. Followed by a rotation of 90° about the o-axis.

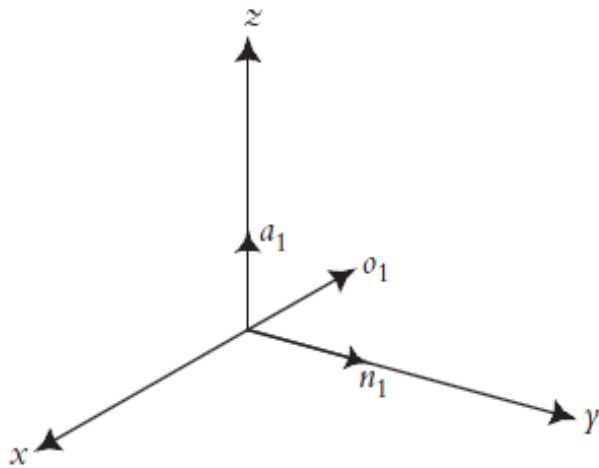
Solution: In this case, since the transformations are made relative to the current frame, each transformation matrix is **post-multiplied**.

$$p_{x,y,z} = Rot(a,90)Trans(4,-3,7)Rot(o,90)p_{noa}$$

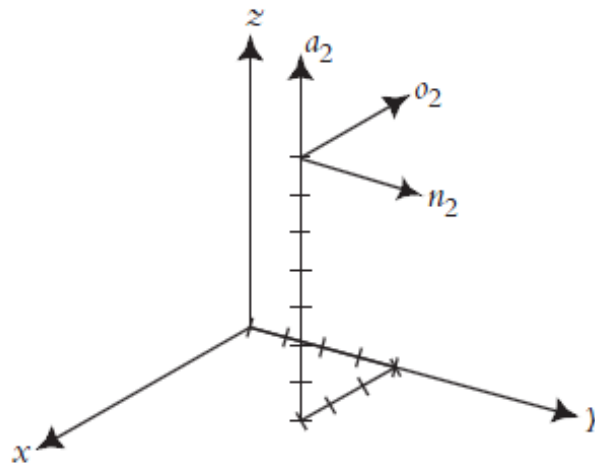
Homogenous Transformation Matrices

• Transformations Relative to the Rotating Frame

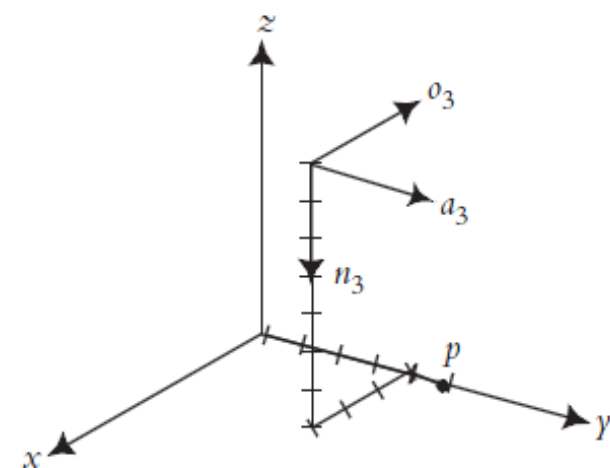
Example (cont'd):



After the 1st transformation



After the 2nd transformation



After the 3rd transformation

$$p_{x,y,z} = Rot(a,90)Trans(4,-3,7)Rot(o,90)p_{noa}$$

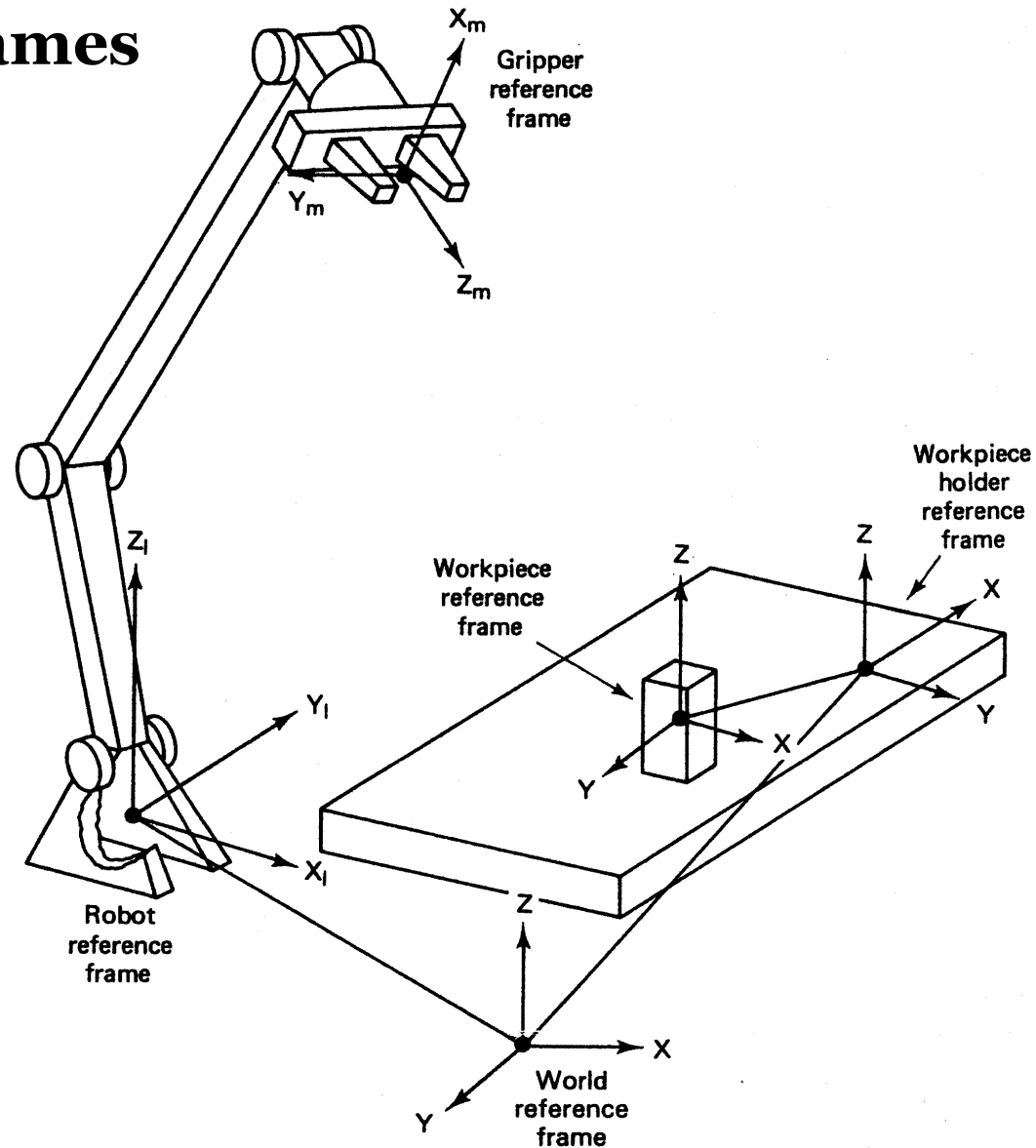
$$= \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 7 \\ 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 0 \\ 1 \end{bmatrix}$$

Outline

- Kinematics
- Robots as Mechanisms
- Matrix Representation
- Homogenous Transformation Matrices
- **Inverse of Transformation Matrices**
- Forward and Inverse Kinematic Equations [For Reading]
- Summary

Inverse of Transformation Matrices

- Reference Frames



Inverse of Transformation Matrices

• Convention

$${}^U p = {}^U T_R \times {}^R p$$

${}^U T_R$: Transformation (reading it as the transformation of frame R relative to frame U (for Universe))

${}^R p$: p relative to frame R

${}^U p$: p relative to frame U

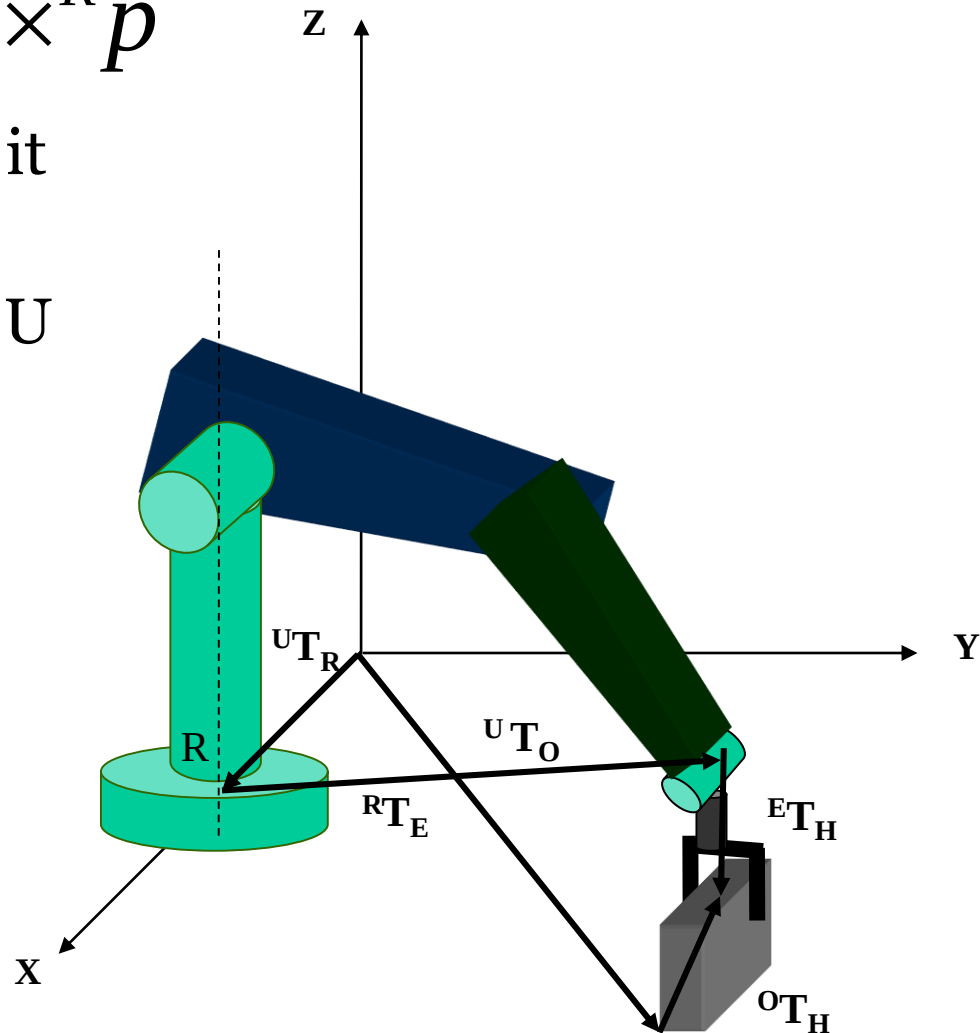
${}^U T_R$: Transformation - robot $\rightarrow$ universe

${}^R T_E$: Transformation - extreme $\rightarrow$ robot

${}^E T_H$: Transformation - hand $\rightarrow$ extreme

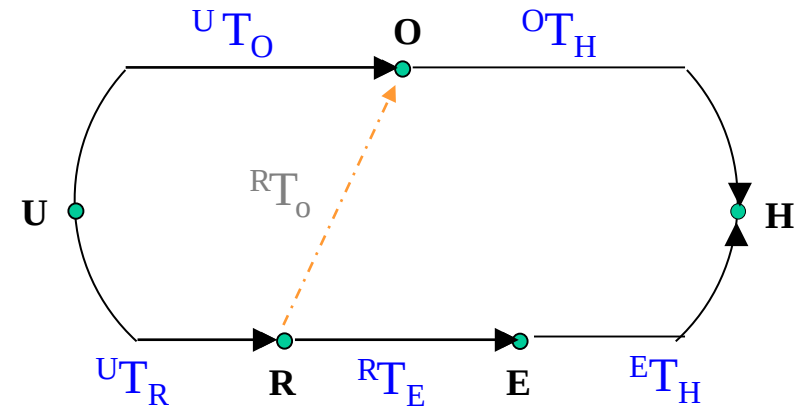
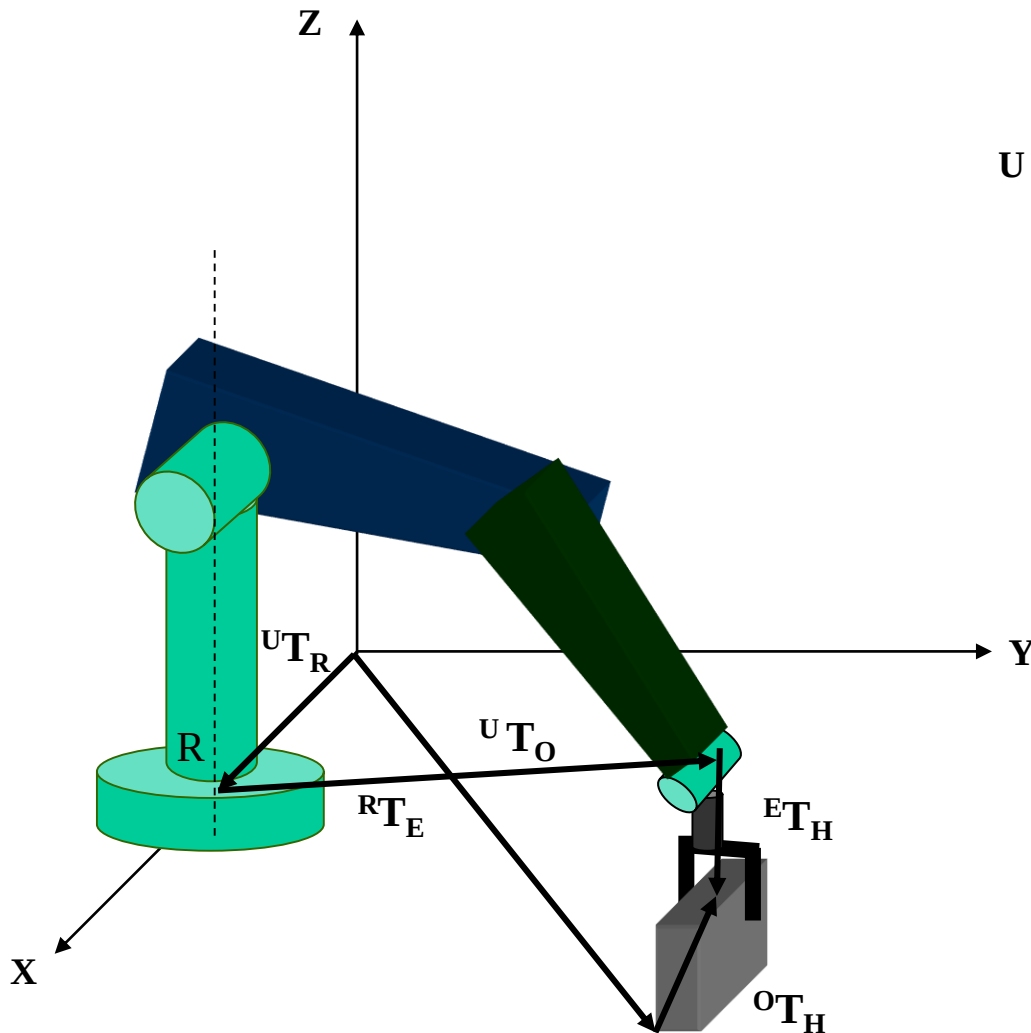
${}^U T_O$: Transformation - object $\rightarrow$ universe

${}^O T_H$: Transformation - hand $\rightarrow$ object

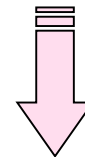


Inverse of Transformation Matrices

• Transform Graph



$${}^U T_R \quad {}^R T_E \quad {}^E T_H = {}^U T_O \quad {}^O T_H$$



$${}^R T_O = {}^R T_E \quad {}^E T_H \quad ({}^O T_H)^{-1}$$

or

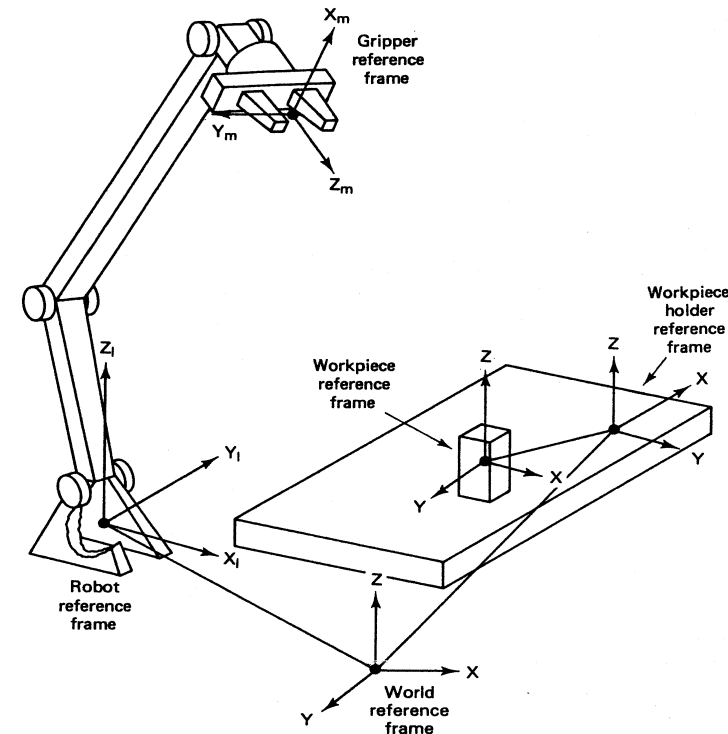
$${}^R T_O = ({}^U T_R)^{-1} \quad {}^U T_O$$

Inverse of Transformation Matrices

• Transform Graph: *Example*

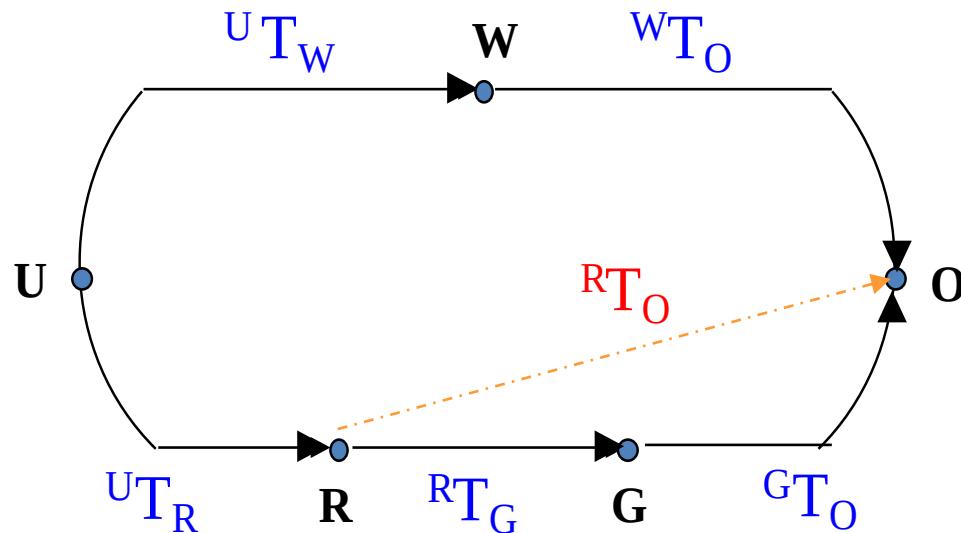
Draw a transform graph for the following arrangement and use it to find ${}^R T_O$, i.e, the transformation of the workpiece reference frame (O) relative to the robot reference frame (R) assuming that the following transformation are known:

- ${}^U T_R$: Transformation of robot reference frame (R) relative to world reference frame (U).
- ${}^R T_G$: Transformation of gripper reference frame (G) relative to robot reference frame (R).
- ${}^G T_O$: Transformation of workpiece reference frame (O) relative to gripper reference frame (G).
- ${}^U T_W$: Transformation of workpiece holder reference frame (W) relative to world reference frame (U).
- ${}^W T_O$: Transformation of workpiece reference frame (O) relative to workpiece holder reference frame (W).



Inverse of Transformation Matrices

- Transform Graph: *Example*



$${}^R T_O = {}^R T_G \times {}^G T_O = ({}^U T_R)^{-1} \times {}^U T_W \times {}^W T_O$$

Inverse of Transformation Matrices

Matrix Inversion: ONLY for homogenous transformation matrices

Assume that **A** is a homogenous 4x4 square matrix that takes the following form

$$\mathbf{A} = \begin{bmatrix} \mathbf{R} & \mathbf{T} \\ \mathbf{0} & \mathbf{1} \end{bmatrix}$$

where **R** is the rotation matrix 3x3, **T** is the translation vector 3x1 and **0** represents the deformation vector 1x3 and **1** is the scale. The inverse can be calculated as follows:

$$\mathbf{A}^{-1} = \begin{bmatrix} \mathbf{R}^T & \mathbf{R}^T \mathbf{x}(-\mathbf{T}) \\ \mathbf{0} & \mathbf{1} \end{bmatrix}$$

where

A⁻¹ is the inverse of **A**,

R^T is the transpose of **R**.

Inverse of Transformation Matrices

Matrix Inversion: Another way to invert a square 4x4 matrix is to assume that an inverse matrix of the following form exists, such that when multiplied by the given matrix, a unit matrix results.

Assume that A is a homogenous 4x4 square matrix and $X=A^{-1}$, then $AX=I$ or

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where the X matrix is the inverse of the A matrix that we are looking for.

Inverse of Transformation Matrices

Matrix Inversion (cont'd):

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note that this represents a set of 12 equations and 12 unknowns.

If you multiply the first matrix A by the first column of the X matrix, the result will be the first column of the unit matrix.

Then there will be a set of 3 equations that you will have to solve for each column.

Inverse of Transformation Matrices

Matrix Inversion - Classical method: the following steps must be taken to calculate the inverse of a matrix:

- Calculate the determinant of the matrix.
- Transpose the matrix.
- Replace each element of the transposed matrix by its own minor (adjoint matrix).
- Divide the converted matrix by the determinant.

Example: Calculate the inverse of the given transformation matrix:

$$T = \begin{bmatrix} 0.5 & 0 & 0.866 & 3 \\ 0.866 & 0 & -0.5 & 2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Sol.:

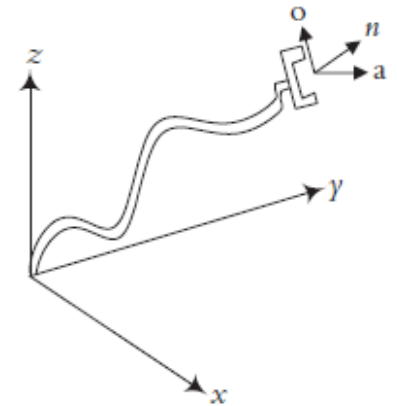
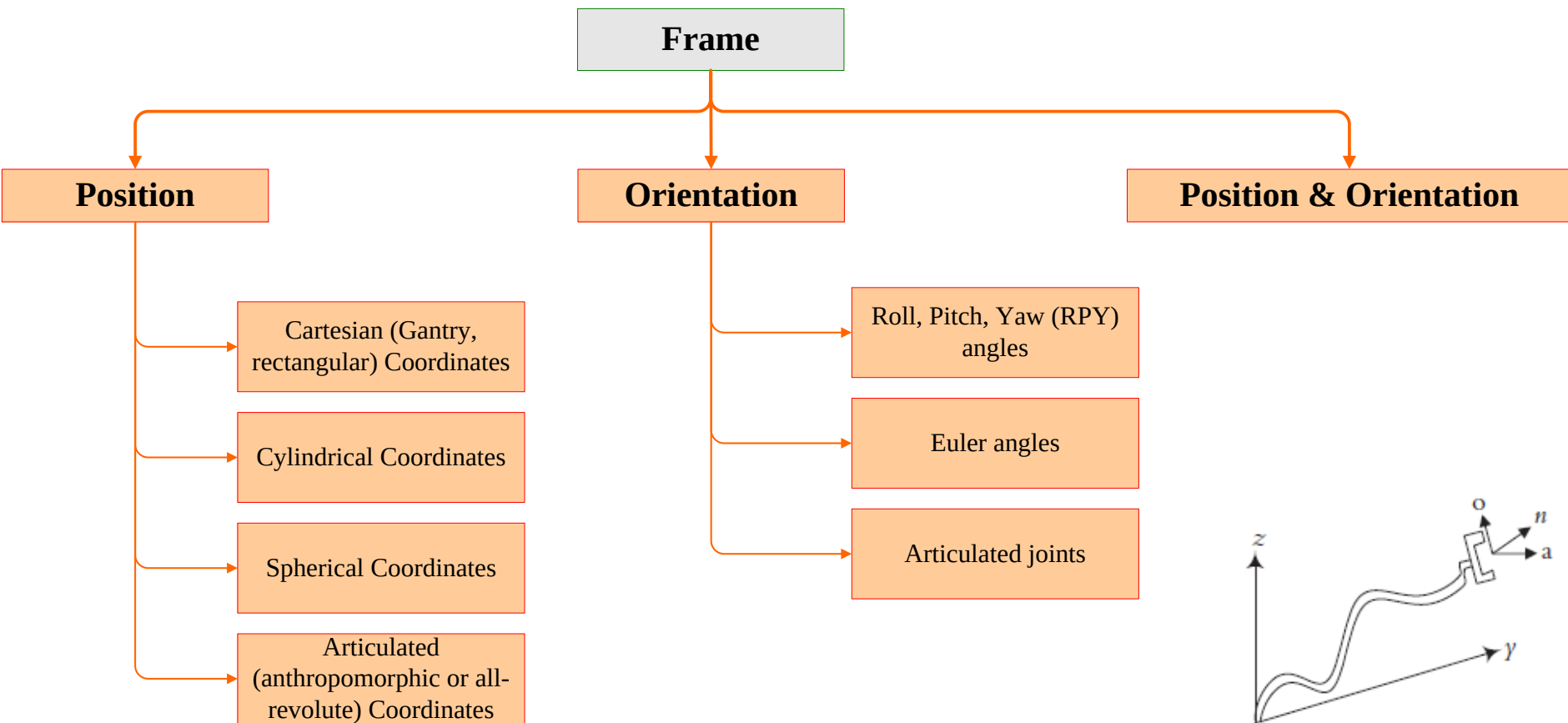
$$\begin{aligned} T^{-1} &= \begin{bmatrix} 0.5 & 0.866 & 0 & -(3 \times 0.5 + 2 \times 0.866 + 5 \times 0) \\ 0 & 0 & 1 & -(3 \times 0 + 2 \times 0 + 5 \times 1) \\ 0.866 & -0.5 & 0 & -(3 \times 0.866 + 2 \times -0.5 + 5 \times 0) \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0.866 & 0 & -3.23 \\ 0 & 0 & 1 & -5 \\ 0.866 & -0.5 & 0 & -1.598 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

To verify: $TT^{-1}=I$

Outline

- Kinematics
- Robots as Mechanisms
- Matrix Representation
- Homogenous Transformation Matrices
- Inverse of Transformation Matrices
- **Forward and Inverse Kinematic Equations** [For Reading]
- Summary

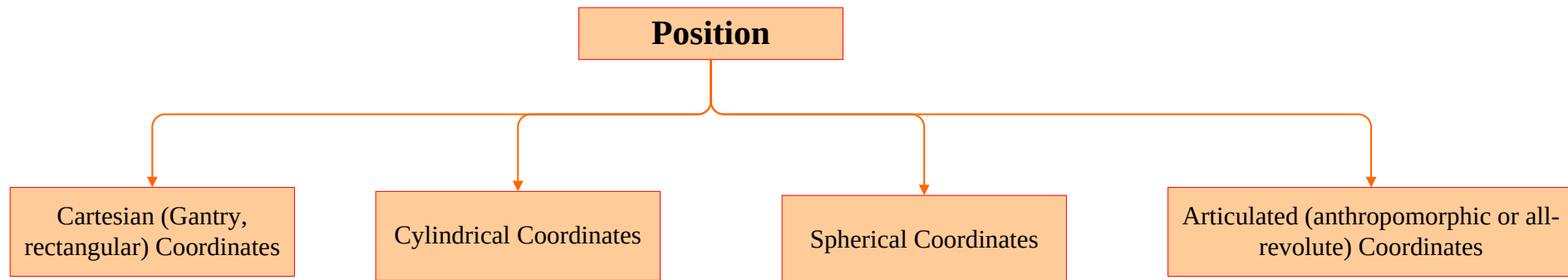
Forward and Inverse Kinematic Equations



The hand frame of the robot relative to the reference frame

Forward and Inverse Kinematic Equations

- **Position:**



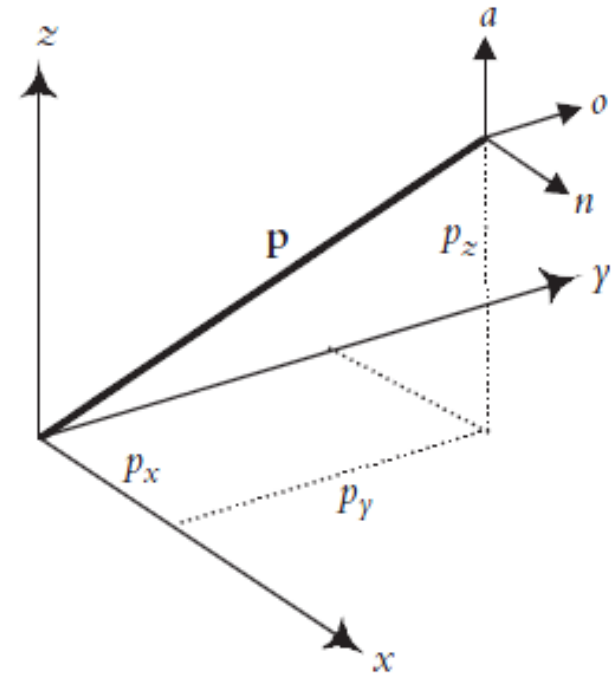
Forward and Inverse Kinematic Equations

- **Position: Cartesian (Gantry, Rectangular) Coordinates**

The transformation matrix representing the forward kinematic equation of the position of the hand of the robot in a Cartesian coordinate system will be:

$${}^R T_p = T_{cart}(p_x, p_y, p_z) = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

For the inverse kinematic solution, simply set the desired position equal to p .



Forward and Inverse Kinematic Equations

- **Position: Cartesian (Gantry, Rectangular) Coordinates**

Example: It is desired to position the origin of the hand frame of a Cartesian robot at point $p = [3, 4, 7]^T$. Calculate the necessary Cartesian coordinate motions that need to be made.

Solution: Setting the forward kinematic equation, represented by the ${}^R T_p$ matrix, equal to the desired position will yield the following result:

$${}^R T_p = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{or} \quad p_x = 3, p_y = 4, p_z = 7$$

Forward and Inverse Kinematic Equations

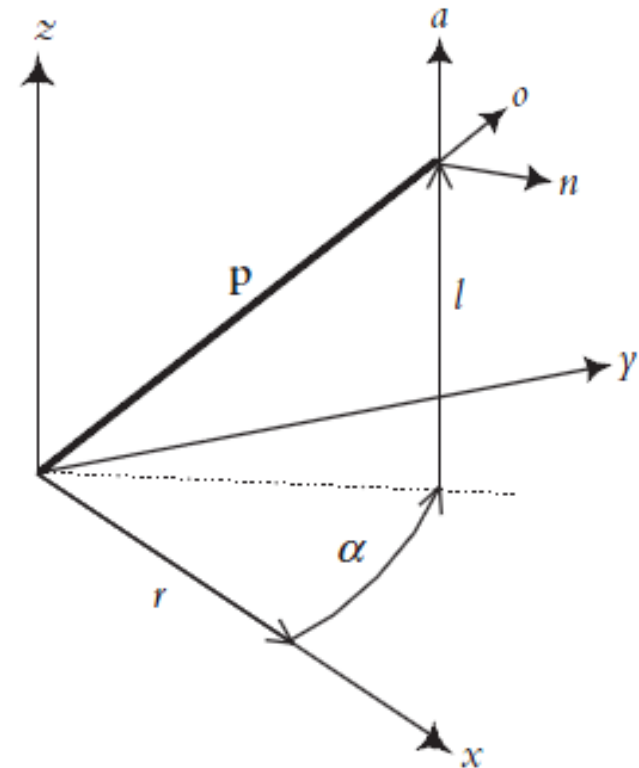
- **Position: Cylindrical Coordinates**

A **cylindrical** coordinate system includes **two linear translations** and **one rotation**. The sequence is a translation of r along the x -axis, a rotation of α about the z -axis, and a translation of l along the z -axis.

$${}^R T_p = T_{cyl}(r, \alpha, l) = Trans(0, 0, l) Rot(z, \alpha) Trans(r, 0, 0)$$

$${}^R T_p = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} C\alpha & -S\alpha & 0 & 0 \\ S\alpha & C\alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & r \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^R T_p = T_{cyl}(r, \alpha, l) = \begin{bmatrix} C\alpha & -S\alpha & 0 & rC\alpha \\ S\alpha & C\alpha & 0 & rS\alpha \\ 0 & 0 & 1 & l \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Forward and Inverse Kinematic Equations

- **Position: Cylindrical Coordinates**

Example: Suppose we desire to place the origin of the hand frame of a cylindrical robot at $[3,4,7]^T$. Calculate the joint variables of the robot.

Solution: Setting the components of the location of the origin of the frame from the T_{cyl} matrix to the desired values, we get:

$$l = 7$$

$$rC\alpha = 3 \quad \text{and} \quad rS\alpha = 4 \quad \text{and therefore, } \tan\alpha = 4/3 \text{ and } \alpha = 53.1^\circ$$

Forward and Inverse Kinematic Equations

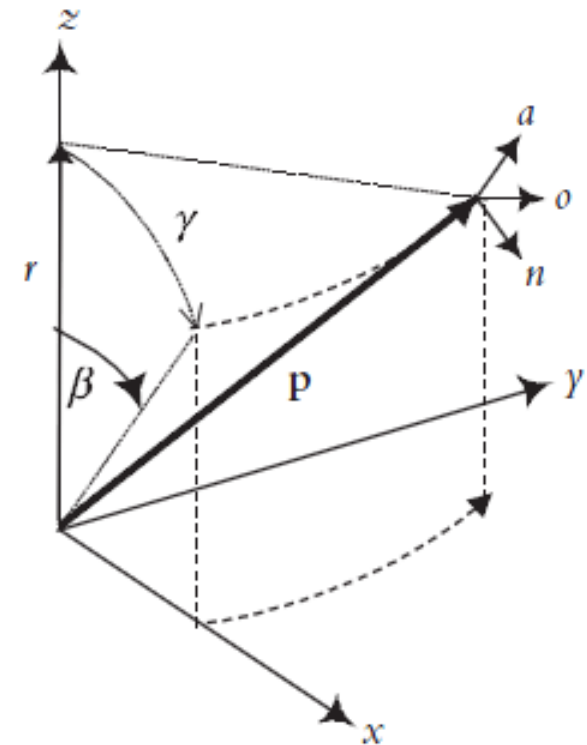
• Position: Spherical Coordinates

A spherical coordinate system consists of **one linear motion** and **two rotations**. The sequence is a translation of r along the z -axis, a rotation of β about the y -axis, and a rotation of γ about the z -axis.

$${}^R T_P = T_{sph}(r, \beta, \gamma) = Rot(z, \gamma) Rot(y, \beta) Trans(0, 0, r)$$

$${}^R T_P = \begin{bmatrix} C\gamma & -S\gamma & 0 & 0 \\ S\gamma & C\gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} C\beta & 0 & S\beta & 0 \\ 0 & 1 & 0 & 0 \\ -S\beta & 0 & C\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & r \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^R T_P = T_{sph}(r, \beta, \gamma) = \begin{bmatrix} C\beta C\gamma & -S\gamma & S\beta C\gamma & rS\beta C\gamma \\ C\beta S\gamma & C\gamma & S\beta S\gamma & rS\beta S\gamma \\ -S\beta & 0 & C\beta & rC\beta \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Forward and Inverse Kinematic Equations

• Position: Spherical Coordinates

Example: Suppose we want to place the origin of the hand of a spherical robot at $[3,4,7]^T$. Calculate the joint variables of the robot.

Solution: Setting the components of the location of the origin of the frame from T_{sph} matrix to the desired values, we get:

$$rS\beta C\gamma = 3$$

$$rS\beta S\gamma = 4$$

$$rC\beta = 7$$

$$\tan\gamma = \frac{4}{3} \rightarrow \gamma = 53.1^\circ \quad \text{or} \quad 233.1^\circ$$

$$\text{then} \quad S\gamma = 0.8 \quad \text{or} \quad -0.8$$

$$\text{and} \quad C\gamma = 0.6 \quad \text{or} \quad -0.6$$

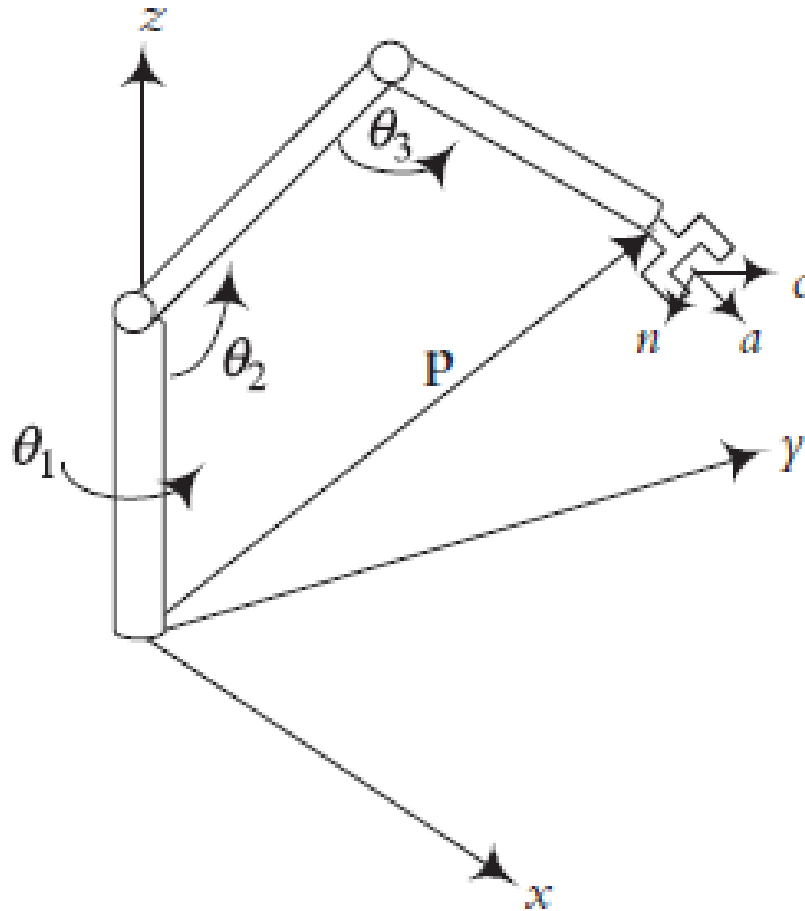
$$\text{and} \quad rS\beta = \frac{3}{0.6} = 5 \quad \text{or} \quad -5$$

$$\text{and since} \quad rC\beta = 7, \rightarrow \beta = 35.5^\circ \quad \text{or} \quad -35.5^\circ$$

$$\text{and} \quad r = 8.6$$

Forward and Inverse Kinematic Equations

- Position: Articulated Coordinates



Articulated coordinates

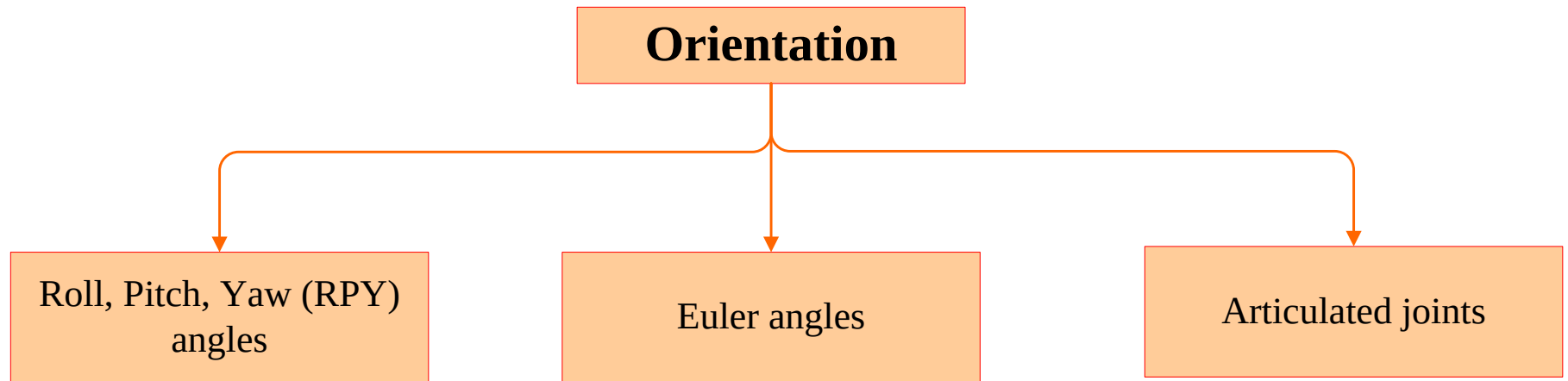
Forward and Inverse Kinematic Equations

- **Position: Articulated Coordinates**

See Denavit-Hartenberg representation (next lecture).

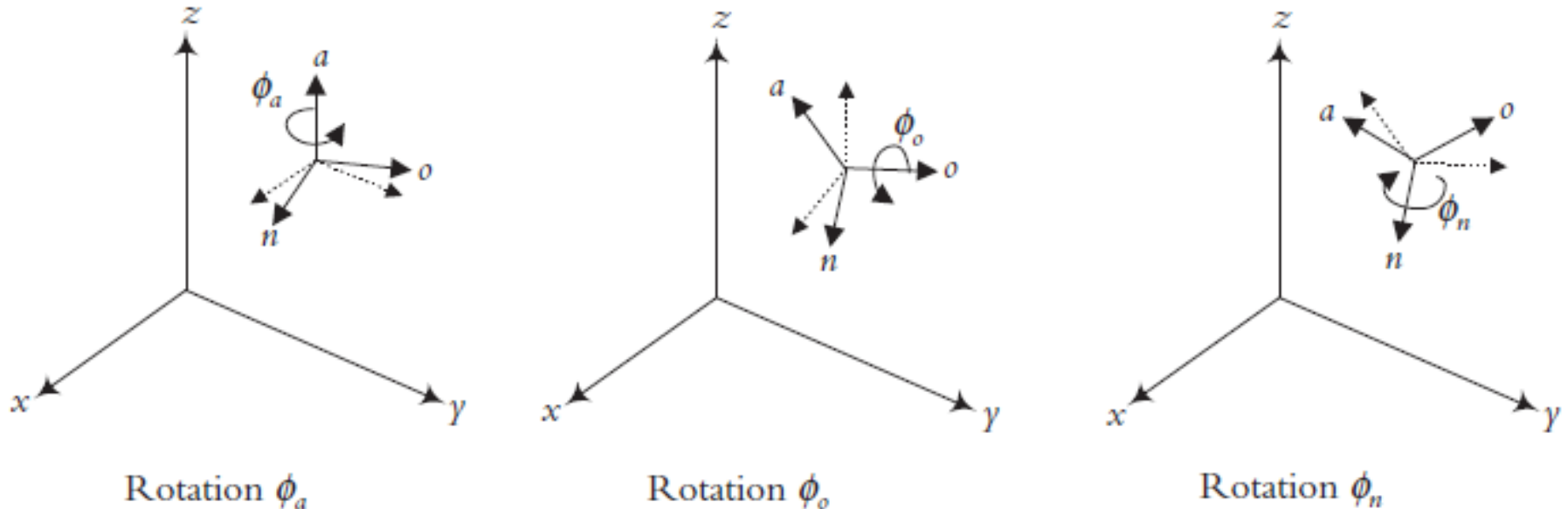
Forward and Inverse Kinematic Equations

- **Rotation:**



Forward and Inverse Kinematic Equations

- **Rotation: Roll, Pitch, Yaw (RPY)**



RPY rotations about the current axes

- Roll: Rotation of ϕ_a about the a-axis (z-axis of the moving frame),
- Pitch: Rotation of ϕ_o about the o-axis (y-axis of the moving frame),
- Yaw: Rotation of ϕ_n about the n-axis (x-axis of the moving frame).

Forward and Inverse Kinematic Equations

- **Rotation: Roll, Pitch, Yaw (RPY)**

The matrix representing the RPY orientation change will be:

$$\text{RPY}(\phi_a, \phi_o, \phi_n) = \text{Rot}(a, \phi_a) \text{Rot}(o, \phi_o) \text{Rot}(n, \phi_n)$$

$$= \begin{bmatrix} C\phi_a C\phi_o & C\phi_a S\phi_o S\phi_n - S\phi_a C\phi_n & C\phi_a S\phi_o C\phi_n + S\phi_a S\phi_n & 0 \\ S\phi_a C\phi_o & S\phi_a S\phi_o S\phi_n + C\phi_a C\phi_n & S\phi_a S\phi_o C\phi_n - C\phi_a S\phi_n & 0 \\ -S\phi_o & C\phi_o S\phi_n & C\phi_o C\phi_n & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Why post-multiplication?

Forward and Inverse Kinematic Equations

- **Rotation: Roll, Pitch, Yaw (RPY)**

Assuming that the final desired orientation achieved by RPY is represented by the (n,o,a) matrix, we will have:

$$\text{RPY}(\phi_a, \phi_o, \phi_n) = \text{Rot}(a, \phi_a) \text{Rot}(o, \phi_o) \text{Rot}(n, \phi_n) = \begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Rot}(a, \phi_a)^{-1} \begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \text{Rot}(o, \phi_o) \text{Rot}(n, \phi_n)$$

Forward and Inverse Kinematic Equations

- **Rotation: Roll, Pitch, Yaw (RPY)**

Multiplying the matrices, we will get:

$$\begin{bmatrix} n_x C\phi_a + n_y S\phi_a & o_x C\phi_a + o_y S\phi_a & a_x C\phi_a + a_y S\phi_a & 0 \\ n_y C\phi_a - n_x S\phi_a & o_y C\phi_a - o_x S\phi_a & a_y C\phi_a - a_x S\phi_a & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\phi_o & S\phi_o S\phi_n & S\phi_o C\phi_n & 0 \\ 0 & C\phi_n & -S\phi_n & 0 \\ -S\phi_o & C\phi_o S\phi_n & C\phi_o C\phi_n & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Forward and Inverse Kinematic Equations

- **Rotation: Roll, Pitch, Yaw (RPY)**

Multiplying the matrices, we will get:

$$\begin{bmatrix} n_x C\phi_a + n_y S\phi_a & o_x C\phi_a + o_y S\phi_a & a_x C\phi_a + a_y S\phi_a & 0 \\ n_y C\phi_a - n_x S\phi_a & o_y C\phi_a - o_x S\phi_a & a_y C\phi_a - a_x S\phi_a & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\phi_o & S\phi_o S\phi_n & S\phi_o C\phi_n & 0 \\ 0 & C\phi_n & -S\phi_n & 0 \\ -S\phi_o & C\phi_o S\phi_n & C\phi_o C\phi_n & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solving these
equations gives:

$$\phi_a = ATAN2(n_y, n_x)$$

$$\phi_o = ATAN2(-n_z, (n_x C\phi_a + n_y S\phi_a))$$

$$\phi_n = ATAN2((-a_y C\phi_a + a_x S\phi_a), (o_y C\phi_a - o_x S\phi_a))$$

Forward and Inverse Kinematic Equations

- **Rotation: Roll, Pitch, Yaw (RPY)**

Note:

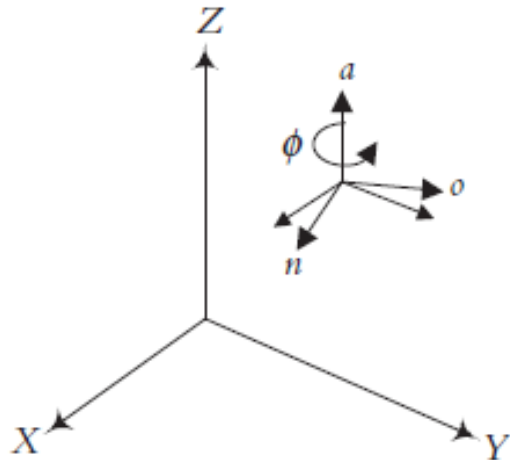
$$\begin{aligned}\phi_a &= ATAN2(n_y, n_x) \\ \phi_o &= ATAN2(-n_z, (n_x C\phi_a + n_y S\phi_a)) \\ \phi_n &= ATAN2((-a_y C\phi_a + a_x S\phi_a), (o_y C\phi_a - o_x S\phi_a))\end{aligned}$$

In terms of the standard arctan function, that is with range of $(-\pi/2, \pi/2)$, two-argument function ATAN2 can be expressed as follows:

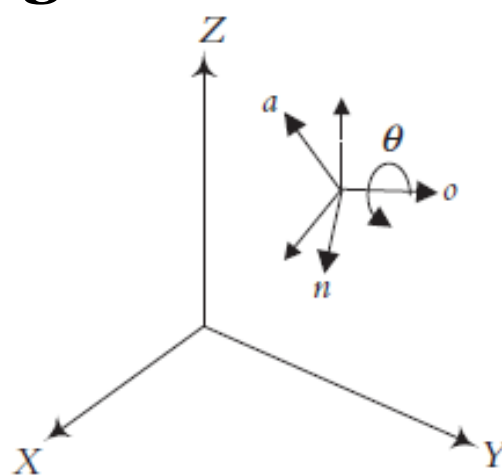
$$\text{atan2}(y, x) = \begin{cases} \arctan(\frac{y}{x}) & x > 0 \\ \pi + \arctan(\frac{y}{x}) & y \geq 0, x < 0 \\ -\pi + \arctan(\frac{y}{x}) & y < 0, x < 0 \\ \frac{\pi}{2} & y > 0, x = 0 \\ -\frac{\pi}{2} & y < 0, x = 0 \\ \text{undefined} & y = 0, x = 0 \end{cases}$$

Forward and Inverse Kinematic Equations

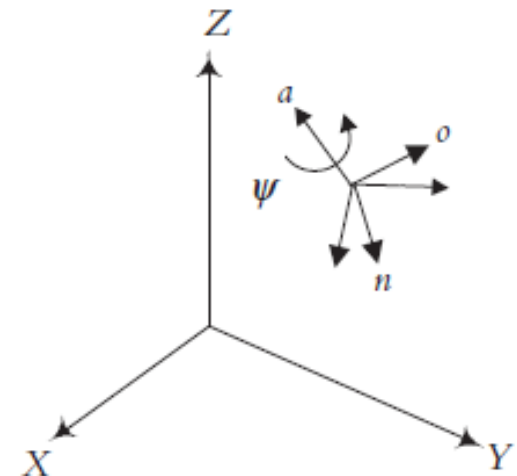
• Rotation: Euler Angles



Rotation of ϕ about the a -axis



Rotation of θ about the o -axis



Rotation of ψ about the a -axis

Euler rotations about the current axes

- Rotation ϕ about the a -axis (z-axis of the moving frame) followed by,
- Rotation of θ about the o -axis (y-axis of the moving frame) followed by,
- Rotation of ψ about the a -axis (z-axis of the moving frame).

Forward and Inverse Kinematic Equations

- **Rotation: Euler Angles**

The matrix representing the Euler angles orientation change will be:

$$\begin{aligned} \text{Euler}(\phi, \theta, \psi) &= \text{Rot}(a, \phi) \text{Rot}(o, \theta), \text{Rot}(a, \psi) \\ &= \begin{bmatrix} C\phi C\theta C\psi - S\phi S\psi & -C\phi C\theta S\psi - S\phi C\psi & C\phi S\theta & 0 \\ S\phi C\theta C\psi + C\phi S\psi & -S\phi C\theta S\psi + C\phi C\psi & S\phi S\theta & 0 \\ -S\theta C\psi & S\theta S\psi & C\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Forward and Inverse Kinematic Equations

• Rotation: Euler Angles

Assuming the final desired orientation achieved by Euler angles is represented by the (n,o,a) matrix:

$$Rot^{-1}(a, \phi) \times \begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} C\theta C\psi & -C\theta S\psi & S\theta & 0 \\ S\psi & C\psi & 0 & 0 \\ -S\theta C\psi & S\theta S\psi & C\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

or,

$$\begin{bmatrix} n_x C\phi + n_y S\phi & o_x C\phi + o_y S\phi & a_x C\phi + a_y S\phi & 0 \\ -n_x S\phi + n_y C\phi & -o_x S\phi + o_y C\phi & -a_x S\phi + a_y C\phi & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\theta C\psi & -C\theta S\psi & S\theta & 0 \\ S\psi & C\psi & 0 & 0 \\ -S\theta C\psi & S\theta S\psi & C\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Forward and Inverse Kinematic Equations

- **Rotation: Euler Angles**

$$\phi = ATAN2(a_y, a_x)$$

$$\psi = ATAN2(-n_x S\phi + n_y C\phi, -o_x S\phi + o_y C\phi)$$

$$\theta = ATAN2(a_x C\phi + a_y S\phi, a_z)$$

Forward and Inverse Kinematic Equations

• Rotation: Euler Angles

Example: The desired final orientation of the hand of a Cartesian-Euler robot is given. Find the necessary Euler angles.

$${}^R T_H = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.579 & -0.548 & -0.604 & 5 \\ 0.540 & 0.813 & -0.220 & 7 \\ 0.611 & -0.199 & 0.766 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution: From the above equations, we find:

$$\phi = ATAN2(a_y, a_x) = ATAN2(-0.220, -0.604) = 20^\circ \text{ or } 200^\circ$$

Realizing that both the sines and cosines of 20° and 200° can be used for the remainder,

$$\psi = ATAN2(-n_x S\phi + n_y C\phi, -o_x S\phi + o_y C\phi) = (0.31, 0.952) = 18^\circ \text{ or } 198^\circ$$

$$\theta = ATAN2(a_x C\phi + a_y S\phi, a_z) = ATAN2(-0.643, 0.766) = -40^\circ \text{ or } 40^\circ$$

Forward and Inverse Kinematic Equations

- **Rotation: Articulated**

Articulated joints consist of three rotations other than the above. See Denavit-Hartenberg representation (next lecture).

Forward and Inverse Kinematic Equations

- **Position and Rotation**

The matrix representing the final location and orientation of the robot is a combination of the above, depending on which coordinates are used. If a robot is made of a Cartesian and an RPY set of joints, then the location and the final orientation of the frame relative to the reference frame will be the product of the two matrices representing the Cartesian position change and the RPY. The robot may be represented by:

$${}^R T_H = T_{cart}(p_x, p_y, p_z) \times \text{RPY}(\phi_a, \phi_o, \phi_n)$$

If the robot is designed based on spherical coordinates for positioning and Euler angles for orientation

$${}^R T_H = T_{sph}(r, \beta, \gamma) \times \text{Euler}(\phi, \theta, \psi)$$

Outline

- Kinematics
- Robots as Mechanisms
- Matrix Representation
- Homogenous Transformation Matrices
- Inverse of Transformation Matrices
- Forward and Inverse Kinematic Equations [For Reading]
- **Summary**

Summary

- Calculating the position and orientation of the hand of the robot is called forward kinematic analysis. In other words, if all robot joint variables are known, using **forward kinematic** equations, we can calculate where the robot is at any instant.
- However, if we want to place the hand of the robot at a desired location and orientation, we need to know how much each link length or joint angle of the robot must be such that—at those values—the hand will be at the desired position and orientation. This is called **inverse kinematic** analysis.